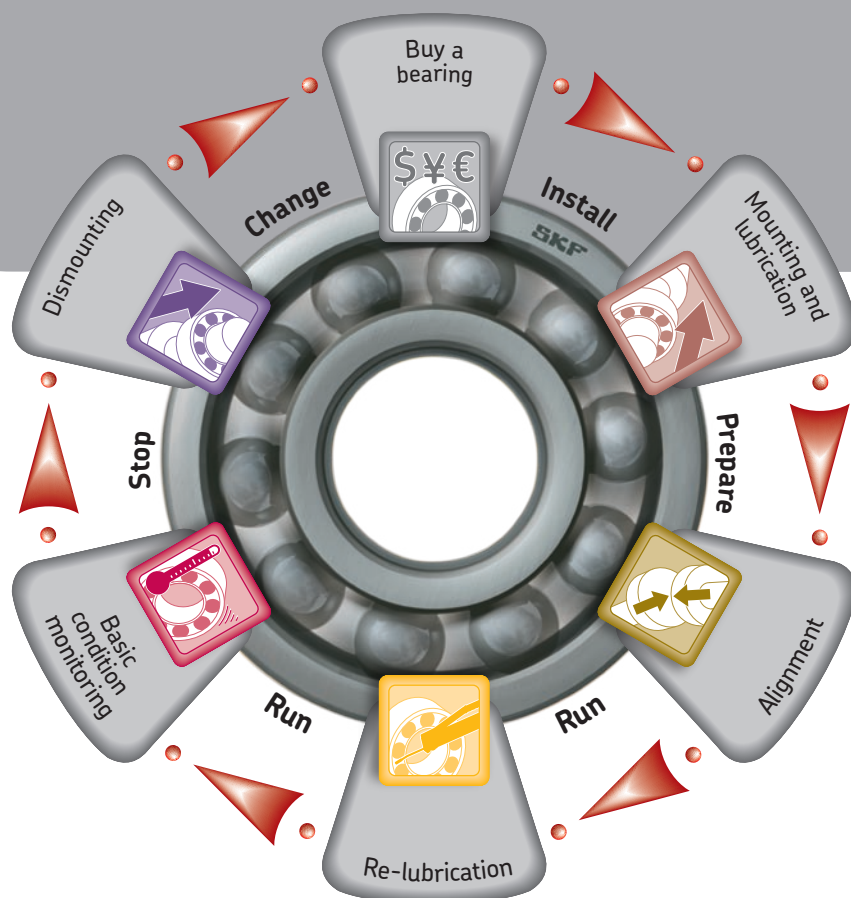
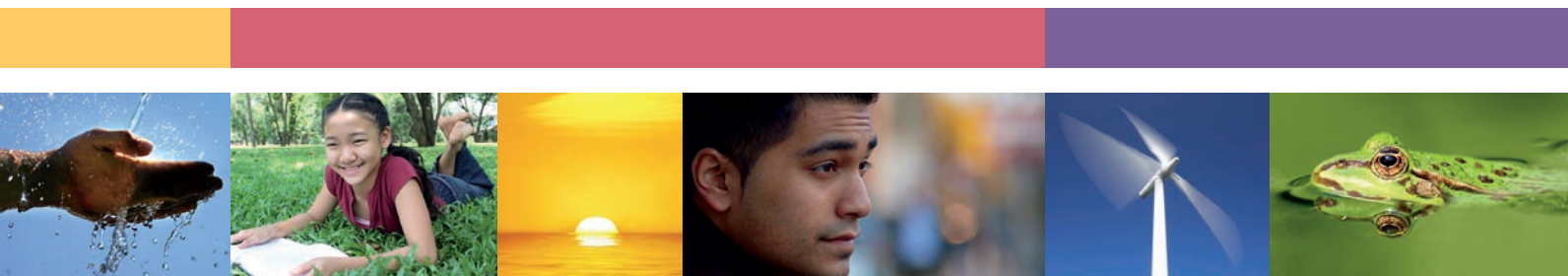




Extending the Bearing Life Cycle

SKF Maintenance and Lubrication Products



SKF Maintenance and Lubrication Products

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“Sustainability for the SKF Group is about combining responsibility to run and develop our business successfully with our responsibility to safeguard resources for future generations.”

The SKF range of Maintenance and Lubrication products has been carefully selected to help achieve maximum bearing service life. For more than 30 years, our main priority has been to provide our customers with solutions to achieve maximum bearing life and consequently optimized machine performance. Of course, safety of machines and operators is of paramount importance and all our products help contribute to this goal.

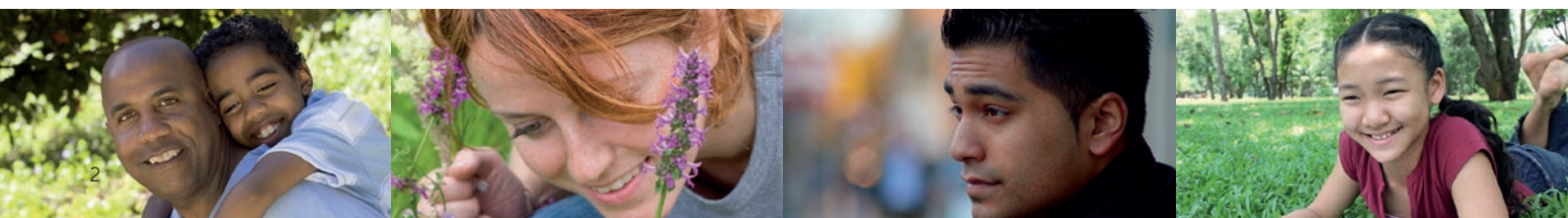
Over the years we have used SKF's extensive knowledge on bearings to develop an efficient programme of tools and methods for mounting and dismounting bearings. When mounting and dismounting bearings, operator safety is our main concern. Therefore, we have developed a wide range of fitting tools and pullers, each ergonomically designed to help minimize the risk of personal injuries and maximize operator safety.

Sometimes bearings are dismantled with the intention to be re-manufactured or re-used. In this case, selecting the correct methods and tools is essential in reducing the risk of bearing damage. Compared to new bearings, each bearing that is re-used results in significant cost savings, extended service life of bearings and machinery, and contributes to environment sustainability by material and energy savings.

When a bearing has been mounted on an application, the application should be aligned. Misalignment results in excessive friction and vibration consequently reducing the bearing and machinery service life. But most important, misalignment increases energy consumption. We advise the use of SKF shaft and belt alignment tools to minimize unnecessary energy consumption.

Using high quality lubricants and correct lubrication methods are key elements to reaching optimum bearing service lifetime. Re-lubrication significantly decreases friction resulting in energy savings. Our wide range of SKF lubricants are specially formulated for a wide range of applications. One of the most special lubricants in our range is the SKF Biodegradable bearing grease, suitable for applications where contamination of the environment is a concern.

SKF automatic lubricators are developed to deliver the right amount of lubricant at the right time to the right place. By doing so, only the required amount of lubricant is used, thus creating less waste and saving on resources. In addition, using SKF lubricators contribute to cleaner workspaces and less hazardous working environments usually caused by oil and grease spillage.



our sfully with enerations”

*Tom Johnstone,
President and CEO, AB SKF*

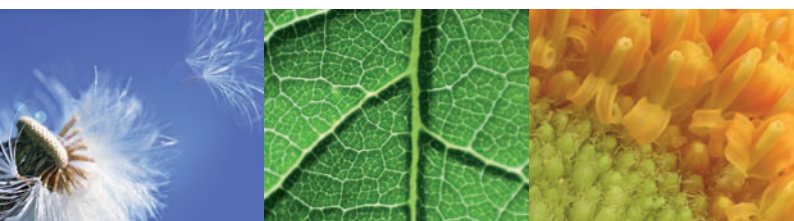


To determine the condition of machinery and bearings we offer a wide range of easy to use instruments. By monitoring machinery condition, such as the operating temperature or speed, the operator can positively adjust the machine's performance thereby helping to reduce energy costs and increase uptime. Many of our instruments are of the non-contact type, contributing to operator safety. This is especially true for machines in hazardous or hard to reach environments.

We aim to constantly further develop our range of products and solutions, finding better ways to save energy and use our scarce resources. This is an ongoing process we practice daily in developing our products.

One example is our range of induction heaters which have been constantly enhanced since SKF innovated the use of them for bearing mounting over 25 years ago. Today, the newest generation of SKF induction heaters reduces energy consumption by up to 80%. This significant achievement is just an example of our commitment to offer sustainable products and services.

At SKF, being environmentally- and ecologically-conscious plays a central role in our day to day business development and this helps us to constantly further develop our range of offerings. For more information about sustainability at SKF, please visit www.skf.com



As part of our responsibility towards saving our natural resources, this publication has been printed on environmentally friendly paper certified by the Forest Stewardship Council. To help us achieve our sustainability goals we kindly request you recycle this publication after use.

The SKF Bearing

Help your bearing achieve its maximum

Every bearing has a pre-calculated service lifetime. However, research has shown that for various reasons, not every bearing achieves it. Important stages, which have a major impact on a bearing service lifetime can be recognised during the bearing's lifecycle. These stages are mounting and lubrication, alignment, re-lubrication, basic condition monitoring and dismantling.

The stages in a bearing life cycle are extremely important for achieving the maximum service life of the bearing.

By applying the right maintenance practices and using the correct tools, you can considerably extend your bearing's service life and increase plant productivity and efficiency.

Mounting and lubrication



Includes mechanical fitting tools, induction heaters and hydraulic equipment

Mounting is one of the critical stages of the bearing's lifecycle. If the bearing is not mounted properly using the correct method and tools, the bearing's service lifetime will be reduced. Lubrication is also an important step in the mounting procedure. Selecting bearing grease suitable for the application is critical to achieving optimum performance. Additionally, the quantity of grease and the lubrication method used can positively influence the service life of the bearing.

Alignment



Includes shaft and belt alignment tools and machinery shims

After the bearing has been mounted in an application, such as a motor connected to a pump, the application should be aligned. If the application is not properly aligned, the misalignment can cause the bearing to suffer additional load, friction and vibration. These can accelerate fatigue and reduce the bearing's, as well as other machine components, service life. Furthermore, increased vibration and friction can significantly increase energy consumption and the risk of premature failures.

Re-lubrication



Includes bearing greases, manual and automatic lubricators and lubrication accessories

When operating, the bearing requires correct re-lubrication practices to optimise its performance. Selecting bearing grease suitable for the application and applying the right quantities at correct intervals are essential for achieving the maximum service life of the bearing. Additionally, the re-lubrication method used can positively contribute to optimising the bearing's service life. Continuous lubrication using automatic lubricators, single or multiple-point, provides more consistent, correct and contamination-free grease supply than manual re-lubrication methods.

Basic condition monitoring

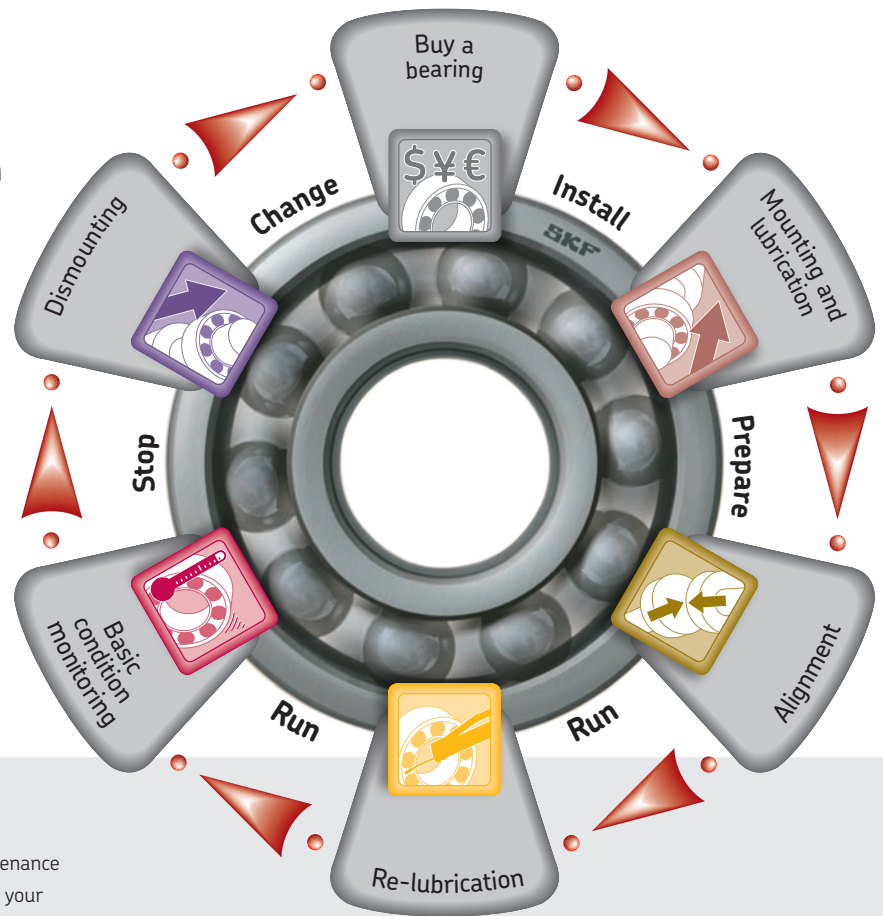


Includes temperature, noise, speed and vibration measuring instruments

During operation, it is important to regularly inspect the condition of the bearing by performing basic condition monitoring, such as temperature, vibration and noise measurements. These regular inspections will allow the detection of potential problems and help to prevent unexpected machine stops. Consequently the machine maintenance can be planned to suit the production schedule, increasing the plant's productivity and efficiency.

Life Cycle

service lifetime



How to use this catalogue

Inside this catalogue you will find SKF's complete range of maintenance products, which can help you get the maximum service life from your bearings. Products included in this catalogue are arranged according to the stages of the bearing life cycle: Mounting and Lubrication, Alignment, Re-lubrication, Basic Condition Monitoring and Dismounting. To help you locate the product you need as easily as possible, we have developed the following quick reference guide:

For more information about SKF maintenance products or to order any of these products, please contact your local SKF authorised distributor or SKF sales company. On the Internet, SKF can be found at www.skf.com. SKF Maintenance Products can be found at www.mapro.skf.com.

Dismounting



Includes pullers, both mechanical and hydraulic, induction heaters and hydraulic equipment

At some point, the bearing will reach the end of its service life and will have to be replaced. Although the bearing may not be used again, it is extremely important to dismount it correctly so that the service life of the replacement bearing is not compromised. Firstly, the use of proper dismounting methods tools will help prevent damage to other machine components, such as the shaft and housing, which are often re-used. Secondly, incorrect dismounting techniques can be hazardous to the operator.



Found next to a product: Indicating that this product is a new addition to the SKF range



Located next to each product: Indicates on which page you can find the technical data and ordering details for that product

Technical data and ordering details: Located on pages 125 – 158 provides the complete list of technical data and ordering details per product

Designation index: Located on pages 159 – 160 lists all products by designation followed by product description in alpha-numerical order

Prevent over 60% of premature bearing failures



16%

Poor fitting

Around 16% of all premature bearing failures are caused by poor fitting (usually brute force...) and being unaware of the availability of the correct fitting tools. Individual installations may require mechanical, hydraulic or heat application methods for correct and efficient mounting or dismounting. SKF offers a complete range of tools and equipment to make these tasks easier, quicker and more cost effective, backed up by a wealth of service engineering know-how. Professional fitting, using specialised tools and techniques, is another positive step towards achieving maximum machine uptime.



36%

Poor lubrication

Although 'sealed-for-life' bearings can be fitted and forgotten, some 36% of premature bearing failures are caused by incorrect specification and inadequate application of the lubricant. Inevitably, any bearing deprived of proper lubrication will fail long before its normal service lifespan. Because bearings are usually the least accessible components of machinery, neglected lubrication frequently compounds the problem. Wherever manual maintenance is not feasible, fully automatic lubrication systems can be specified by SKF for optimum lubrication. Effective lubrication, using only recommended SKF greases, tools and techniques, helps to significantly reduce downtime.



14%

Contamination

A bearing is a precision component that will not operate efficiently unless both the bearing and its lubricants are isolated from contamination. And, since sealed-for-life bearings in ready-greased variants account for only a small proportion of all bearings in use, at least 14% of all premature bearing failures are attributed to contamination problems. SKF has an unrivalled bearing manufacturing and design capability and can tailor sealing solutions for the most arduous operating environments.



34%

Fatigue

Whenever machines are overloaded, incorrectly serviced or neglected, bearings suffer from the consequences, resulting in 34% of all premature bearing failures. Sudden or unexpected failure can be avoided, since neglected or overstressed bearings emit 'early warning' signals, which can be detected and interpreted using SKF condition monitoring equipment. The SKF range includes hand-held instruments, hard-wired systems and data management software for periodic or continuous monitoring of key operating parameters.



Mounting and lubrication

Mounting

Mounting bearings in a cold condition

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Interference fits: tapered shafts	
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Bearing Fitting Tool Kit TMFT 36	
Hook Spanners HN series	
Adjustable Hook Spanners HNA series	
Impact Spanners TMFN series	
Hook Spanners HN ../SNL series	
Bearing Lock Nut Spanner TMHN 7 series	
Axial Lock Nut Sockets TMFS series	
Bearing Handling Tool TMMH series	

Mounting bearings using heat

High Frequency Portable Induction Heater TMBH 1	
Electric Hot Plate 729659 C	
Small Induction Heater TIH 030m	
Medium Induction Heater TIH 100m	
Large Induction Heater TIH 220m	
Induction Heater Trolley TIH T1	
Special heaters for large components	

Mounting bearings using hydraulic techniques

SKF Oil Injection Method	
Oil Injection Method CD-ROM	
SKF Drive-up Method	
Hydraulic Nut Drive-up Adapter HMVA 42/200	
SKF Drive-up Method CD-ROM	
Adapter and withdrawal sleeves for oil injection	
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Mounting and lubrication

Mounting

Prevent 16% of premature bearing failures

Around 16% of all premature bearing failures are a result of poor fitting or using incorrect mounting techniques. Individual applications may require mechanical, heat or hydraulic mounting methods for correct and efficient bearing mounting. Selecting the mounting technique appropriate for your application will help you extend your bearing's service life and reduce costs resulting from premature bearing failure as well as potential damage to the application.

Mounting bearings in a cold condition

Small and medium size bearings are generally cold mounted. Traditionally the bearing is mounted using a hammer and a length of old pipe. This practice can cause forces to be transmitted through the rolling elements, causing damage to the raceways. SKF fitting tools help prevent bearing damage by applying the forces to the bearing ring with the interference fit.

Mounting bearings using heat

Oil baths are often used for heating bearings prior to mounting. However, this method can contaminate the bearing, resulting in premature bearing failure. Today, induction heating is the most common technique for heating bearings since it allows a high degree of controllability, efficiency and safety. SKF has set the standards for the development of induction heaters for bearing applications. SKF bearing induction heaters are equipped with many features, which help prevent bearing damage during heating.

Mounting bearings using hydraulic techniques

SKF has pioneered the use of hydraulic techniques, such as the SKF Oil Injection Method and the SKF Drive-up Method, for mounting bearings. These techniques have helped to simplify bearing arrangements and facilitate correct and easy mounting. SKF has also developed a comprehensive range of tools and equipment to put these hydraulic techniques into effect.

Online mounting and dismounting instructions

At skf.com/mount, SKF offers a unique Web-based, free of charge information service for the mounting and dismounting of SKF bearings and bearing housings in 13 languages. This service provides step-by-step instructions for mounting or dismounting. The system also provides information on proper tools and lubricants. With this free Internet based service; SKF's expertise is at your fingertips around the clock worldwide.



Mounting bearings in a cold condition

Premature bearing failure can result from damage incurred when a bearing is incorrectly mounted

Typical problems that can cause premature failures are:

- Damage caused during the fitting procedure
- Incorrect sized shafts and housings i.e. too loose or too tight
- Retaining lock nuts coming loose in operation
- Burred and damaged shaft and housing seats and shoulders
- Incorrectly mounted bearings

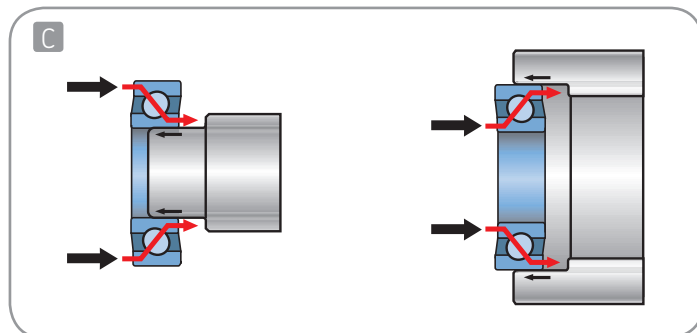
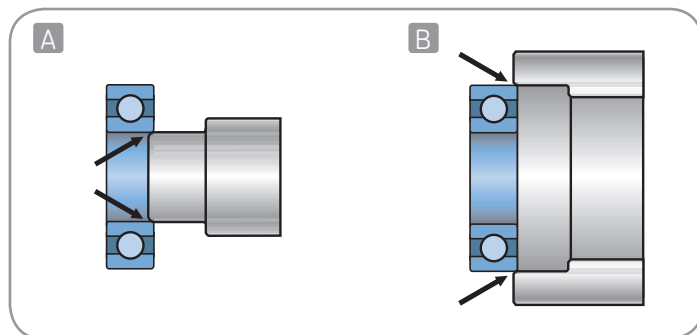
Interference fits: cylindrical shafts

Most bearings are fitted to their shaft or housing with one component having an interference fit. For determining the correct fit, refer to the SKF General Catalogue, the SKF Maintenance Handbook or consult an SKF application engineer.



Incorrect mounting

When bearings are mounted cold, care must be taken to ensure the drive-up forces are applied to the ring with the interference fit. Damage and a resulting bearing failure can occur if the mounting force is transmitted through the rolling elements causing damage to the raceways.



Correct mounting

The correct way to minimise raceway damage is to use the specifically designed tools from SKF, such as the Bearing Fitting Tool Kit TMFT 36. These tools allow drive-up forces to be applied effectively and evenly to the component with the interference fit avoiding raceway damage.

- A Shaft interference fit
- B Housing interference fit
- C Uneven distribution of forces can result in raceway damage
- D With the correct tools raceway damage is avoided



Mounting and lubrication



Interference fits: tapered shafts

Bearings mounted on tapered seatings achieve their interference fit by being driven up the tapered shaft. Care should be taken to ensure the bearing is not

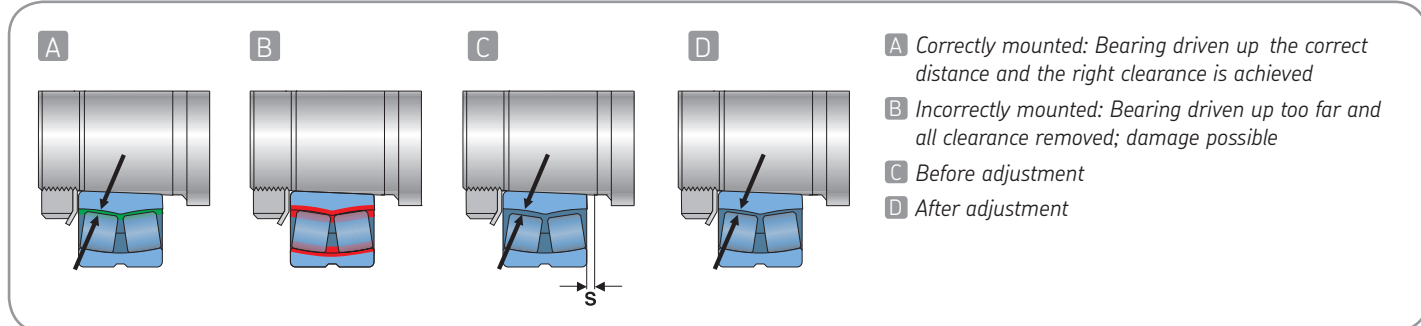
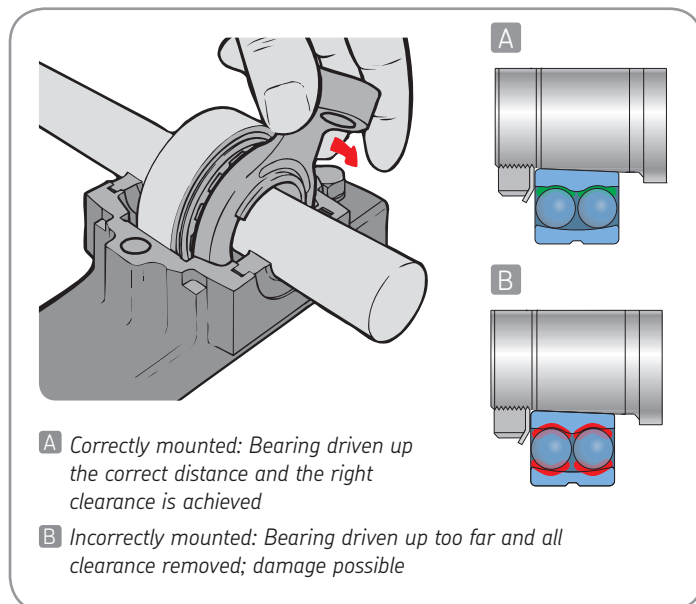
driven up too far, as all the internal clearance may be removed and damage to the bearing is possible.

Spherical roller bearings

Method: Correct adjustment of spherical roller bearings is determined by measuring the residual internal clearance in the bearing or by the amount of axial drive-up. Details of the required reduction of clearance and axial drive-up can be obtained from tables published in the SKF General Catalogue. For larger size bearings, it is generally recommended to consider using a tapered seating to facilitate easy mounting and dismounting.

Self-aligning ball bearings

Method: Adjustment of double row, self-aligning ball bearings is more difficult to achieve than spherical roller bearings because the feeler gauge method cannot be used. A very effective method to mount this type of bearing correctly is to use the SKF TMHN 7 lock nut spanner set.



Anti-fretting Agent LGAF 3E

LGAF 3E is a greasy, smooth paste especially developed to prevent fretting corrosion between metal surfaces in loose fit arrangements. Fretting corrosion is caused by very slight oscillations or by vibrations, which may lead to serious

damage in bearings and other machine parts and can make dismounting almost impossible.

- Reduction of fretting corrosion providing easier dismounting of bearings
- Better sliding on designed loose bearing arrangements such as vibrating screens, truck and car wheel bearings
- Easier removal of general industrial components in a wide range of applications such as nuts, bolts, flanges, studs, bearings, guide pins, couplings, jack screws, lathe centres, push rods, and spline shafts



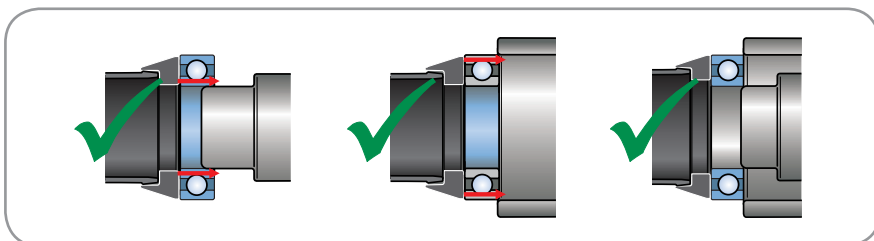
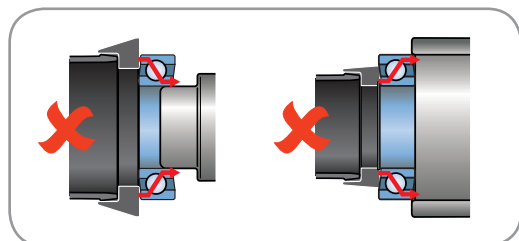
Bearing Fitting Tool Kit TMFT 36

Helps prevent premature bearing failure

Poor fitting, usually using brute force, accounts for 16% of premature bearing failures. The SKF Bearing Fitting Tool Kit TMFT 36 is designed for quick and precise mounting of bearings, while minimising the risk of bearing damage. The right combination of impact ring and sleeve allows effective transmission of mounting force to the bearing ring with the interference fit, minimising the risk of damaging the

bearing's raceways or rolling elements. The kit contains 36 impact rings, 3 impact sleeves and a dead-blow hammer packed in a lightweight carrying case. In addition to mounting bearings, the TMFT 36 is also suitable for mounting other components such as bushings, seals and pulleys.

- 36 impact rings in different sizes facilitate the mounting of more than 400 different bearings
- Facilitates correct mounting on shaft, housing and blind applications
- The diameter of the impact ring perfectly fits the inner and outer diameter of the bearing
- Small diameter of the impact area on top of the sleeve allows effective transmission and distribution of mounting force
- Impact rings and sleeves are made of high-impact resistant material for longevity
- Click connection between impact ring and sleeve provides stability and durability
- The impact rings are suitable for use under a press
- Impact rings are marked for clear visual identification of the ring's size and easy selection
- Even surface of the impact sleeve's body provides excellent grip
- The nylon double-side head of the dead-blow hammer helps to prevent damaging the components
- The rubber handgrip of the dead-blow hammer provides excellent grip

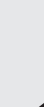




Mounting and lubrication

TMFT 36 selection table

SKF bearing series

SKF bearing series												
Sleeve	Rings	60.. 63.. 16.. 63/.. 62.. 64.. 62/.. 98..	622.. 623.. 630..	12.. 13.. 22.. 23.. 72.. 73..	72.. 73.. 22.. 23.. 72.. 73..	32.. 33.. 52.. 53..	213.. 222.. 223..	10.. 2.. 22.. 3.. 23..	30.. 32.. 31.. 33..	C22.. C40..	42.. 43..	
	10 / 26	629 16100 6000 6200	63000	129								
	10 / 30											
	10 / 35 12 / 28	6300 6001 16101 6201	62300 63001									
	12 / 32											
	12 / 37	6301	62301	1201 2201 1301 2301	7201	3201 5201					4201	
	15 / 32	16002 6002 6202	63002									
	15 / 35											
	15 / 42	6302	62302	1202 2202 1302 2302	7202	3202 5202 3302 5302		202			4202	
	17 / 35	16003 6003 98203 6203 6303	63003									
	17 / 40											
17 / 47												
	20 / 42	16004 98204 6004 6204	63004								32004	
	20 / 47											
	20 / 52	6304	62304	1204 2204 1304 2304	7204	3204 5204 3304 5304	22205/20	204 2204 304 2304 1005	30204		4204	
	25 / 47	16005 6005 62/22 98205 6205 63/22 6305 6403	63005									
	25 / 52											
	25 / 62											
	30 / 55	16006 6006 62/28 98206 6206 63/28 6306 6404	63006									
	30 / 62											
	30 / 72											
	35 / 62	16007 6007 6207	63007									
	35 / 72											
	35 / 80	6307 6405	62307	1207 2207 1307 2307	7207	3207 5207 3307 5307	22207 B52-2207 21307	207 2207 33207 30307 31307 32307	30207 32207 33207 30307 31307 32307	C 2207	4207	
		40 / 68	16008 6008 6208	63008								
		40 / 80										
40 / 90		6308 6406	62308	1308 2308	7308	3308 5308	21308 22308	308 2308	32008/38 30208 32208 33208 32307/37 30308 31308 32308 32009	C 2208	4208	
45 / 75		16009 6009 6209	63009									
45 / 85												
45 / 100		6309 6407	62309	1309 2309	7309	3309 5309	21309 22309	309 2309	30209 32209 33209 358 X 30309 31309 32309 32010 33010 JLM 104948	C 2209	4209	
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50 / 90		6210	62210	1210 2210	7210	3210 5210	22210 B52-2210	210 2210	30210 32210 33210 JM 205149	C 2210	4210	
50 / 110		6310 6408	62310	1310 2310	7310	3310 5310	21310 22310	310 2310	30310 31310 32310 32011 33011		4310	
55 / 90		16011 6011 6211	62211	1211 2211	7211	3211 5211	22211 B52-2211	211 2211	30211 32211 33211 30311 31311 32311	C 2211	4211	
55 / 120		6311 6409	62311	1311 2311	7311	3311 5311	21311 22311	311 2311	30311 31311 32311		4311	

Hook Spanners HN series

Exact spanner radius reduces the risk of nut damage

The HN series includes 15 different size hook spanners based on the DIN 1810 standard. The hook spanners are designed for use with SKF KM nuts as well as any other KM nuts conforming to the DIN 981 standard. Additionally, they are suitable for use with N, AN, KMK, KMFE and KMT as well as nuts manufactured according to the DIN 1804 standard.

- Minimises the risk of shaft and nut damage
- Plastic handle is oil, grease and dirt resistant to provide a better grip
- The plastic handle minimises direct metal to skin contact, reducing the risk of corrosion in the handle area
- Hole in the spanner's handle facilitates easy storage
- Designation of spanner representing its size is laser-engraved allowing easy identification and selection
- Available as a set: HN 4-16/SET containing 9 spanners for lock nut sizes 4 up to 16



Adjustable Hook Spanners HNA series

Four sizes for tightening or loosening up to 24 nut sizes

The SKF Adjustable Hook Spanners HNA series facilitate the easy and safe tightening and loosening of KM, KML, N, AN, KMK, KMFE and KMT nuts. The spanners are made of special hardened steel for durability.

- One hook spanner covers several nut sizes, making it suitable for use with many applications
- Economic solution: 4 hook spanners cover 24 nut sizes
- Laser engraved designation, which represents the range of nut sizes covered by the spanner, allows easy selection of the correct spanner
- Versatile: suitable for KM, KML, N, AN, KMK, KMFE and KMT nuts
- Hole in the spanner's handle facilitates easy storage
- Minimises the risk of shaft and nut damage

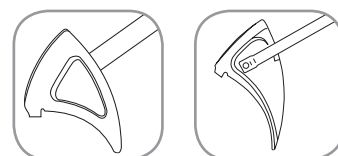


Impact Spanners TMFN series

High impact forces without nut damage

SKF Impact Spanners are designed for safe and easy tightening and loosening of locknuts used to secure and adjust larger bearings directly on the shaft or with adapter and withdrawal sleeves.

- Avoids shaft and nut damage
- Safe and user friendly
- Impact applied effectively to the nut
- Suitable for nuts of series KM, KML, HM..T, HML..T, HM 30, HM 31, AN.., N.. and N... (for nut sizes 23 and above)
- Special wide impact face





Mounting and lubrication

Hook Spanners HN ../SNL series

Easy and quick bearing mounting and dismounting in SNL housings

A normal design HN hook spanner cannot be used in a SKF SNL housing, however the hook spanners of HN ../SNL series are especially designed to facilitate easy and quick mounting and dismounting of bearings with tapered bore on adapter sleeve in SKF SNL bearing housings. They are also suitable for tightening and loosening a wide variety of locknuts in both housing and shaft applications.

- Unique, special design allows the HN ../SNL series to be used inside SKF SNL and SNH bearing housings
- Suitable for tightening and loosening KM, KML, N, AN, KMK, KMFE and KMT lock nuts, facilitating the use in a wide range of housing and shaft applications
- The large contact area of the spanner around the nut provides excellent grip and force transmission
- Exact fit reduces the risk of shaft, nut and housing damage
- Designation is laser-engraved on the handle allowing easy identification and selection
- Additional five larger sizes for nut outer diameter 155 to 210 mm (6.1 to 8.3 in) are available upon request
- Hole in the spanner's handle facilitates easy storage

The HN ../SNL series consists of 16 sizes suitable for nut outer diameter ranging from 38 to 145 mm (1,5 to 5,7 in). The spanners are made of hardened high quality chrome vanadium steel for durability.

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Bearing Lock Nut Spanner TMHN 7 series

For achieving the correct radial clearance

The TMHN 7 set of lock nut spanners is especially designed for mounting self-aligning ball bearings as well as small spherical roller and CARB toroidal roller bearings on tapered seatings. Using the TMHN 7 minimises the risk of over-tightening of the lock nut, which can result in removing the bearing's radial clearance and bearing damage.

- 7 different-sized spanners to fit nut sizes 5 to 11
- Each spanner is clearly marked with correct tightening angle and protractor
- 4 grip points on each spanner giving better and safer grip on the nut
- Reduced risk of damaging bearing by over-tightening
- Suitable for use with lock nuts of the KM series either on shaft or in SNL housings

125

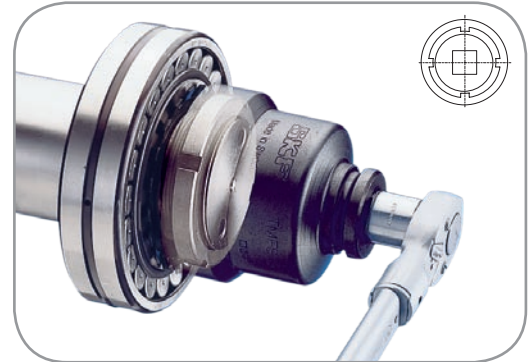


Axial Lock Nut Sockets TMFS series

Easy mounting and dismounting without nut damage

SKF Axial Lock Nut Sockets are designed for safe and easy tightening and loosening of lock nuts. They are used to secure and adjust bearings on tapered journals, adapter sleeves and withdrawal sleeves.

- Demands less space around the bearing arrangement than hook spanners
- Inch connections for power tools or torque wrenches
- TMFS fits nuts of series KM, KMK (metric) and KMF



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Bearing Handling Tool TMMH series

Get a safe grip on handling bearings

The SKF Bearing Handling Tool is a simple, yet ingenious solution to problems associated with handling medium and large size bearings, weighing up to 500 kg (1 100 lb).

The Bearing Handling Tool consists of a steel band with two handles and two anti-rotation plates, which fits around the outer ring of the bearing whilst the bearing is still in horizontal position.

By turning the two handles, the bearing handling tool is tightly fitted around the bearing. The two anti-rotation plates fix the inner ring and the rolling elements, preventing them from swivelling. This combination, the bearing and the bearing handling tool, can then be lifted manually or by a crane and turned to the vertical position safely, easily and quickly.

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Bearing handling has never been safer, easier or quicker

A Place the bearing handling tool around the bearing while it is still in horizontal position.

- One tool suitable for many bearing types and sizes
- Tightly fits around the outer ring
- The two anti-rotation plates fix the inner ring and the rolling elements, preventing them from swivelling during lifting

B Lift the combination, bearing and Bearing handling tool, using a crane.

- The bearing can be lifted from its horizontal position, safely and easily
- The tightly secured bearing is prevented from falling, minimising injury to the operator or damage to itself
- Full surface contact during lifting prevents damage to the bearing, which can be caused by one-point grip or lifting hooks

C Turn the combination to vertical position for placement on the shaft.

- Fixing the inner ring allows easy placement on the shaft and helps preventing damage to the ring or the rolling elements
- Easy and simple, one operator can complete the job

D The bearing is placed on the shaft during mounting.

- The job is safely, easily and quickly done
- Time-savings compared to conventional handling methods can be more than 50%





Mounting bearings using heat

The force needed to mount a bearing increases rapidly with bearing size. Because of the mounting force required, larger bearings cannot easily be pressed onto a shaft or into a housing. Therefore the bearing or the housing is heated before mounting.

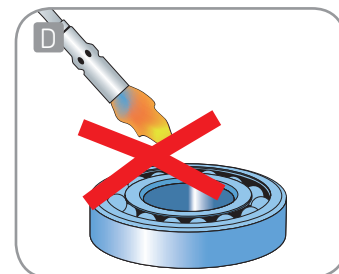
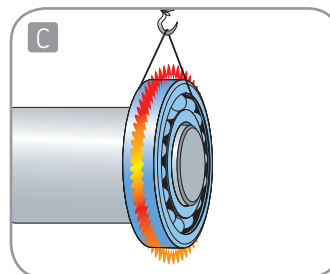
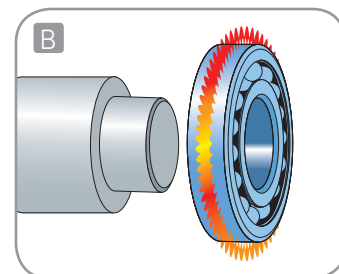
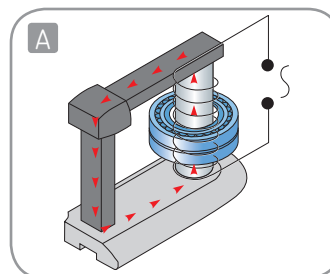
Principle of induction heating

An induction heater can be compared to a transformer using the principle of a primary coil with a large number of windings, and a secondary coil with a few windings, on a mutual iron core. The input/output voltage ratio is equal to the ratio of the windings, while the energy remains the same. Consequently, the secondary coil will provide a low voltage at a high amperage. In the case of the SKF induction heater, the bearing is a short circuited, single turn, secondary coil through which a low A.C. voltage flows at high amperage, thus generating high heat. The heater itself, as well as the yoke, remains at ambient temperature. As this type of heating induces an electric current, the bearing will become magnetised.

It is important to ensure that the bearing is then demagnetised so that it will not attract metal particles during operation. All SKF induction heaters have automatic demagnetising cycles.

Hot mounting

The temperature difference between the bearing and seating depends on the magnitude of the interference fit and the bearing size. Normally a bearing temperature of 80 to 90 °C (144 to 162 °F) above that of the shaft is sufficient for mounting. Never heat a bearing to a temperature greater than 125 °C (257 °F), because the material may change metallurgically and produce alterations in diameter or hardness. Local overheating must be avoided and in particular never heat a bearing using an open flame.



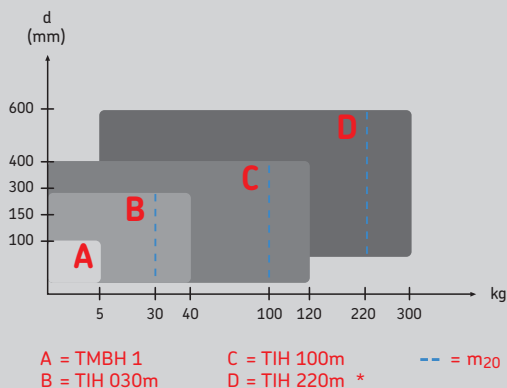
A Principle of induction heating

B Hot mounting

C Lifting gear

D Never heat a bearing using an open flame

Wear clean protective gloves when mounting a hot bearing. Lifting (hoisting) gear can facilitate mounting. Push the bearing along the shaft as far as the abutment and hold the bearing in position, pressing until a tight fit is obtained. SKF supplies a full range of heating tools, such as induction heaters and electric hot plates with an adjustable thermostat and cover for all common mounting needs.



Selection guide

There are no totally restrictive guidelines to follow when choosing your SKF bearing heater. It will depend upon the type and geometrical dimensions of the components you want to heat. Nevertheless, SKF offers the following helpful general selection guide.

SKF m_{20} concept

" m_{20} " represents the weight (kg) of the heaviest SRB 231 bearing which can be heated from 20 to 110 °C (68 to 230 °F) in 20 minutes. This defines the heater's power output instead of its power consumption.

High Frequency Portable Induction Heater TMBH 1

A portable bearing heater weighing only 4,5 kg (10 lb)

The SKF High Frequency Portable Induction Heater TMBH 1 is a portable lightweight heater for heating bearings with an inner diameter ranging from 20 to 100 mm (0.8 to 4 in) and a maximum corresponding weight of 4,5 kg (10 lb). The heater uses a patented method of heating based on high frequency induction, which provides optimised efficiency.

- Lightweight and portable (4,5 kg to 10 lbs)
- Heating efficiency better than 85%
- Components are not magnetised
- Equipped with temperature and time control
- Supplied with a heating clamp, temperature probe, power cable, heat resistant gloves and a carrying case

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Electric Hot Plate 729659 C

Thermostat controlled bearing heating

The SKF Electric Hot Plate, 729659 C, is a professional heating device especially designed for pre-heating small bearings prior to mounting. The temperature of the plate can be adjusted at the turn of a knob to provide a temperature range of between 50 and 200 °C (120 and 390 °F).

- Adjustable temperature range of 50 to 200 °C (120 to 390 °F)
- Protective cover to avoid contamination during heating

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Mounting and lubrication



Small Induction Heater TIH 030m

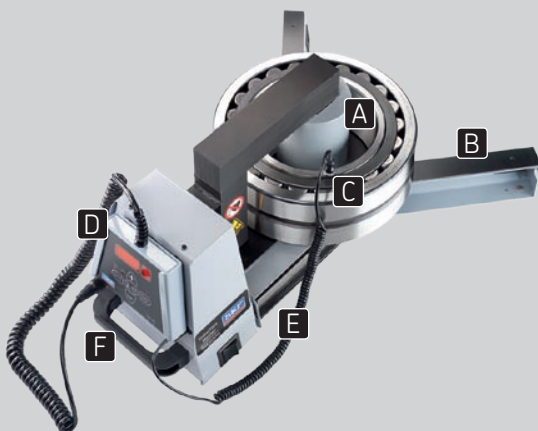
Small bearing heater with high heating capacity of up to 40 kg bearing

The new SKF Small Induction Heater TIH 030m combines high heating capacity with portability. The compact lightweight design makes the TIH 030m portable. Placing the induction coil outside the heater's housing allows the heating of bearings weighing up to 40 kg (88 lb). The heater is equipped with thermal overheating protection to reduce the risk of damage to the induction coil and the electronics.

- Compact lightweight design; just 20,9 kg (46,0 lb) facilitating portability
- 2-step power setting and smaller yokes allow heating smaller bearings safely and at lower power consumption
- Capable of heating a 28 kg (61,7 lb) bearing in just 20 minutes
- Temperature mode pre-set at 110 °C (230 °F) to help prevent bearing over-heating
- Automatic demagnetisation
- 3 years warranty

In addition to temperature mode, the TIH 030m is equipped with a time mode for heating components other than bearings. The heater is supplied standard with three yokes and is available in two executions: 230V/50-60Hz and 100-110V/50-60Hz.

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- A** Induction coil outside the heater's housing allows shorter heating time and lower energy consumption
- B** Foldable bearing support arms facilitate the heating of larger diameter bearings
- C** Magnetic temperature probe helps prevent bearing overheating
- D** Easy-to-use control panel and LED display integrated in a remote control
- E** Internal storage for all 3 yokes reduces the risk of yoke damage or loss
- F** Integrated carrying handle facilitates portability

Medium Induction Heater TIH 100m

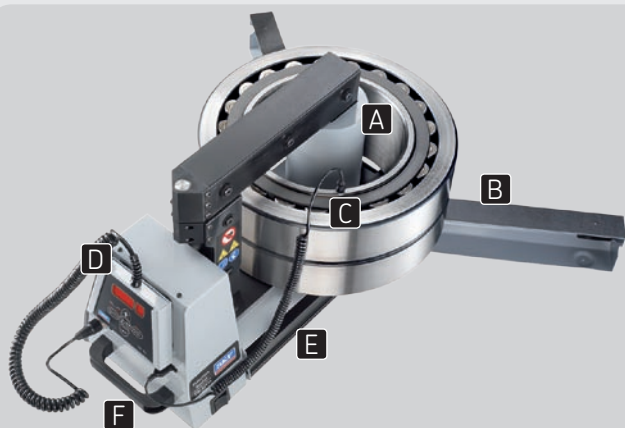
Medium bearing heater with high heating capacity of up to 120 kg bearing

The SKF Medium Induction Heater TIH 100m has the same high standards of efficiency and performance as the small heater combined with increased capacity. The advanced design of the power electronics allows features of accurate electric current control, cut-outs to avoid overheating, controls on rate of temperature increase, these are as some of the standard features in the TIH...m range.

Placing the induction coil outside the heater's housing allows the heating of bearings weighing up to 120 kg (264 lb). The heater is equipped with thermal overheating protection to reduce the risk of damage to the induction coil and the electronics.

- Standard swivel arm for large size yoke
- Capable of heating a 97 kg (213 lb) bearing in less than 20 minutes, saving time and energy
- 2-step power setting and smaller yokes allow heating smaller bearings safely and at lower power consumption
- Temperature mode pre-set at 110 °C (230 °F) to help prevent bearing over-heating
- Automatic demagnetisation
- 3 years warranty

In addition to temperature mode, the TIH 100m is equipped with a time mode for heating components other than bearings. The heater is supplied standard with three yokes and is available in two executions: 230V/50-60Hz or 400-460V/50-60Hz.



- A** Induction coil outside the heater's housing allows shorter heating time and lower energy consumption
- B** Foldable bearing support arms facilitate the heating of larger diameter bearings
- C** Magnetic temperature probe helps prevent bearing overheating
- D** Easy-to-use control panel and LED display integrated in a remote control
- E** Internal storage for all 3 yokes reduces the risk of yoke damage or loss
- F** Integrated carrying handle facilitates portability



Mounting and lubrication



Large Induction Heater TIH 220m

Large bearing heater with high heating capacity of up to 300 kg bearing

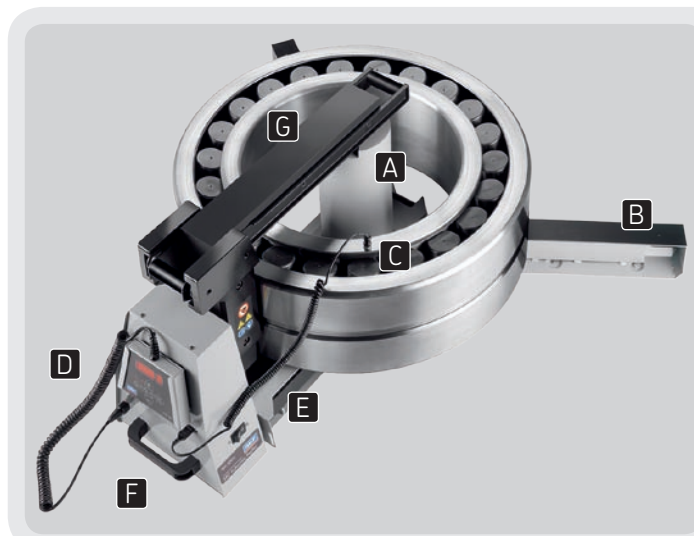
NEW

The SKF Large Induction Heater TIH 220m is a reliable and robust induction heater from the TIH...m range suitable for heating bearings up to a maximum weight of 300 kg (660 lbs) and solid components up to a maximum weight of 150 kg (330 lbs).

Maximum efficiency is achieved by placing the induction coil outside the heater's housing, in the centre of the bearing, resulting in shorter heating time and lower energy consumption. The LED operating display and control panel are integrated in a remote control, which makes the heater easy to use. To suit different operating voltages worldwide, the TIH 220m is available in different voltage variants.

- Capable of heating a 220 kg (480 lbs) bearing in just 20 minutes, saving time and energy
- Supplied standard with two yokes, allowing bearings with a bore diameter from 60 mm (2,3 in) up to a maximum weight of 300 kg (660 lbs) to be heated
- 2-step power setting and smaller yoke facilitates efficient heating of smaller bearings with a lower power consumption
- Temperature mode pre-set at 110 °C (230 °F), helps prevent bearing overheating
- Automatic demagnetisation
- 3 year warranty

Advanced design of the power electronics including current and overheating control, combined with user friendly features such as sliding arms and remote control are standard to the TIH...m range.



- A Induction coil outside the heater's housing allows shorter heating time and lower energy consumption
- B Foldable bearing support arms facilitate the heating of larger diameter bearings
- C Magnetic temperature probe helps prevent bearing overheating
- D Easy-to-use control panel and LED display integrated in a remote control
- E Internal yoke storage for the smaller yoke reduces the risk of yoke damage or loss
- F Integrated carrying handles provide excellent grip when moving the TIH 220m around
- G Sliding arm allows easy and quick bearing replacement

Induction Heater Trolley TIH T1

Move induction heaters from one job to another easily and quickly

The SKF TIH T1 trolley is designed to improve mobility when using SKF induction heaters, especially the larger ones. The trolley has a high carrying capacity of up to 900 kg (1 934 lb) and is fitted with a drawer with an oil resistant mat and two adjustable dividers.

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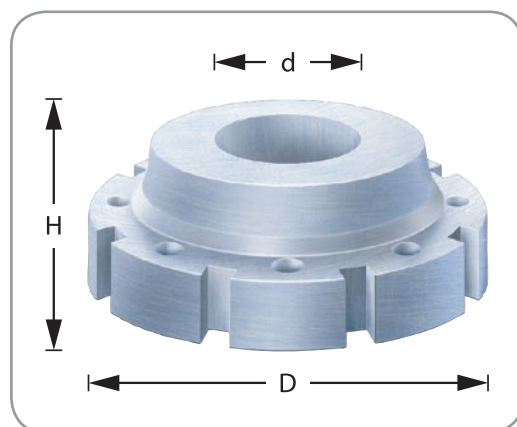


Special heaters for large components

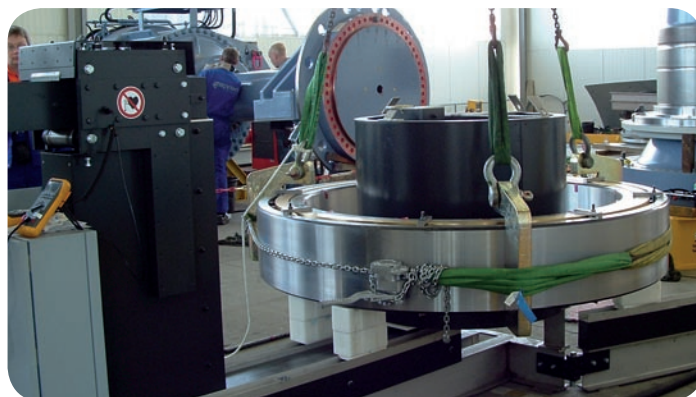
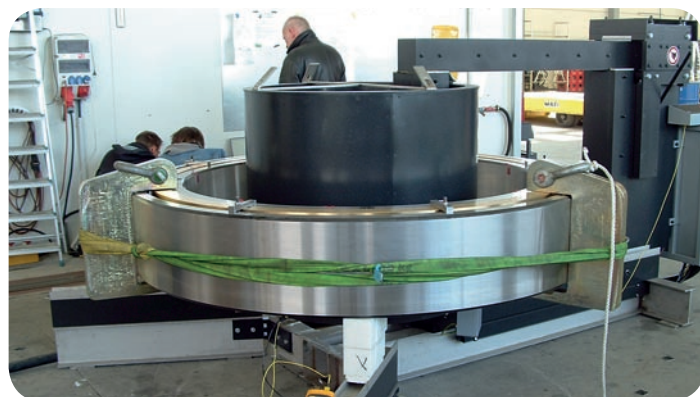
Custom-made to your specifications

SKF can quote for the supply of special heaters for large heating jobs. To provide an accurate quotation the following information is required:

- Dimensions of the component to be heated ($d \times D \times H$)
- Weight in kg or lb
- Required heating temperature
- Desired heating times
- Available power supply
- Demagnetisation requirements
- Temperature or time control requirements
- Portability requirements, if any
- A sketch or drawing of the component to be heated



SKF also offers a range of heating equipment, which can be used for both mounting and dismantling bearings. The range includes aluminium rings, TMBR series, as well as fixed and adjustable EAZ induction heaters. For details of these products, please see pages 115 - 116 of this catalogue.





Mounting bearings using hydraulic techniques

SKF pioneered hydraulic mounting techniques

SKF invented hydraulic techniques for mounting bearings in the 1940 s. Since then, the SKF hydraulic methods have been further developed to become the preferred mounting methods for larger bearings as well as other components. These techniques have helped to simplify bearing arrangements and facilitate correct and easy mounting.

With the SKF hydraulic mounting techniques you can achieve:

- More control, allowing precision, accuracy and repeatability to be maintained
- Minimum risk of damaging the bearings and shafts
- Less manual effort
- Greater operator safety

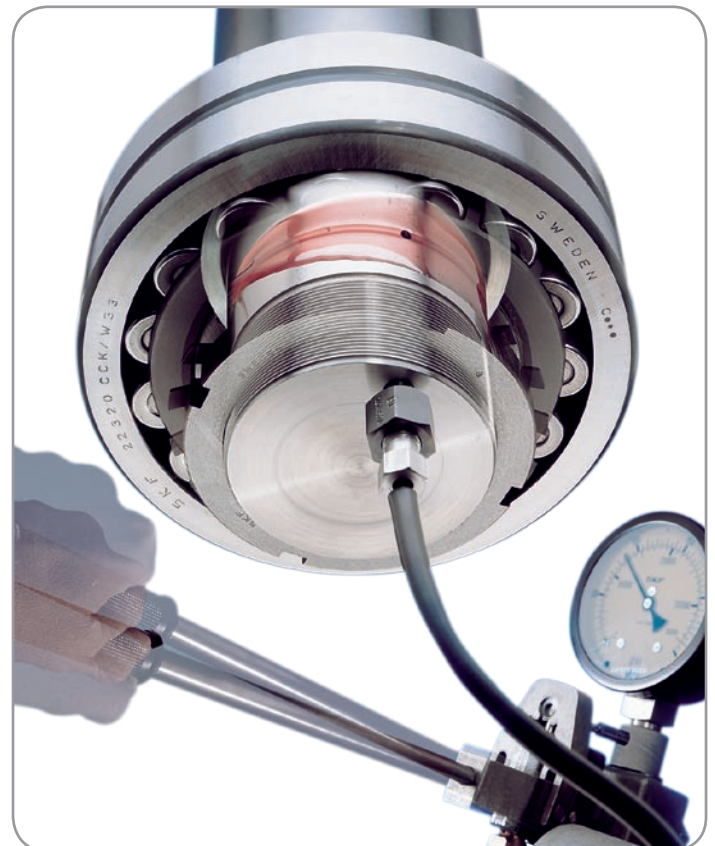
SKF Oil Injection Method

Makes bearing mounting an easy task

The SKF Oil Injection Method allows bearings and other components with an interference fit to be fitted in a safe, controllable and rapid manner. The method does not require keyways to be machined on the shaft, saving valuable time and money in materials and production. Interference fits (also known as shrink fits) have long been recognised for their reliability in transmitting large torsional loads. Very often interference fits offer the only solution when connecting hubs to shafts with intermittent or fluctuating loads.

The SKF Oil Injection Method is used to mount bearings on tapered seatings in combination with a hydraulic nut. The method, which is used for many bearing applications, can also be found in other applications, such as:

- Couplings
- Gear wheels
- Railway wheels
- Propellers
- Built-up crankshafts



Tapered shafts

The concept

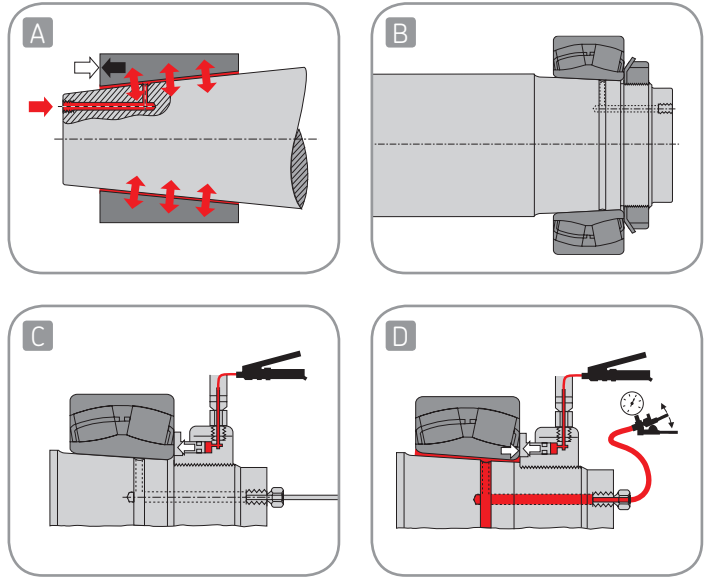
A Injecting oil between two tapered surfaces creates a thin oil film, which reduces the friction between them, thereby significantly reducing the mounting force required. The thin oil film also minimises the risk of metallic contact when mounting, reducing the risk of component damage.

The preparation

B During manufacture the shafts are prepared with oil ducts and grooves. For technical information on how to prepare the shafts, consult an SKF application engineer.

The action

C Bearings are mounted by pushing them up the shaft with the aid of an SKF HMV .. E nut.
D The force to mount the bearing is reduced if oil is injected between the shaft and the bearing. This is often done with larger size bearings.



In addition to mounting bearings on tapered seatings, the SKF Oil Injection Method can be also used for dismounting bearings mounted on either tapered or cylindrical seatings. See page 117 of this catalogue for more details.

Oil Injection Method CD-ROM

The SKF oil injection calculations made easy

The CD-ROM calculation program computes easily the laborious manual calculations often necessary for the SKF Oil Injection Method. Additionally, the CD-ROM provides theoretical details behind the method plus information on designing components, practical experiences, application examples and more. The CD-ROM provides you with detailed instructions and practical information on how to use the SKF Oil Injection

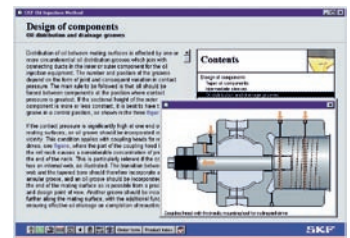
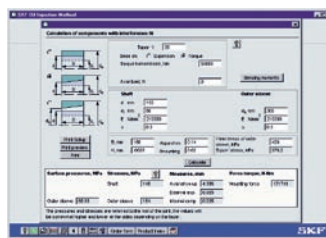
Method for mounting and dismounting bearings, as well as using the method in design, calculation and application of shrink fitted components. In addition, the program includes animations, photographs, detailed product information and instructions for use, as well as video clips showing various methods and techniques.

The CD-ROM is a powerful tool, which includes the following features:

- User-friendly calculation program to determine pressures, stresses and interference levels
- Explanations of the theoretical background
- Information on designing components
- Information on SKF products, which enable the Oil Injection Method to be used
- Practical experiences and application examples
- Complete SKF Drive-up Method program for fitting spherical roller bearings and CARB bearings in a safe, rapid and controlled manner
- Information on related SKF products such as gauges, heaters and pullers

The benefits of using the SKF Oil Injection Method CD-ROM include:

- Substantial time and cost-savings
- Elimination of arithmetic errors
- Ability to see the effects of design changes in seconds
- All information on Oil Injection gathered on one CD-ROM
- Quick and easy access to all the advantages of the Oil Injection Method





Mounting and lubrication

SKF Drive-up Method

Accurate axial drive-up of spherical roller and CARB bearings



The SKF Drive-up Method is a well-proven method of accurately achieving the adjustment of spherical roller and CARB bearings, mounted on tapered seatings, which is unique to SKF. The correct fit is achieved by controlling the axial drive-up of the bearing from a predetermined position.

The method incorporates the use of an SKF HMV ..E hydraulic nut fitted with a dial indicator, and a high accuracy digital pressure gauge, mounted on the selected pump.

- Reduces the use of feeler gauges
- Greatly reduces the time to mount spherical roller and CARB bearings
- A reliable and accurate method of adjustment
- The only suitable way to mount sealed spherical roller and CARB bearings

Special hydraulic pressure tables have been developed, providing the required pressures, for each bearing type. This enables accurate positioning of the bearing at the starting point from where the axial drive-up is measured.



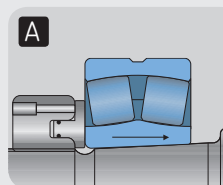
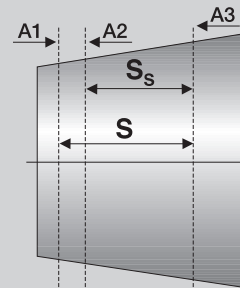
Step by step procedure

1. Ensure that the bearing size is equal to the HMV ..E-nut. (Otherwise the pressure in the table must be adjusted.)
2. Determine whether one or two surfaces slide during mounting; see figures A-D.
3. Lightly oil all mating surfaces with a thin oil, e.g. SKF LHM 300, and carefully place the bearing on the shaft.
4. Drive the bearing up to the starting position by applying the HMV ..E-nut pressure found in the table. Monitor the pressure by the gauge on the selected pump. SKF hydraulic pump 729124 SRB is suitable for hydraulic nuts $\leq$ HMV 54E. SKF TMJL 100SRB is suitable for hydraulic nuts $\leq$ HMV 92E while TMJL 50SRB is suitable for nuts $\leq$ HMV 200E. As an alternative the SKF digital pressure gauge TMJG 100D can be screwed directly into the hydraulic nut.

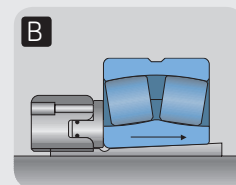
5. Drive the bearing up the taper by the required distance S_s . The axial drive-up is best monitored by a dial indicator. The SKF hydraulic nut HMV ..E is prepared for dial indicators. Normally, the bearing is now mounted with a suitable interference on the shaft and a suitable residual clearance.

For abnormal operating conditions, hollow shafts, very accurate requirements on residual clearance etc., the drive-up must be adjusted. In such cases please contact SKF or refer to the SKF Drive-up Method CD-ROM or skf.com/mount.

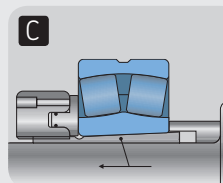
A1 Zero position
A2 Starting position
A3 Final position



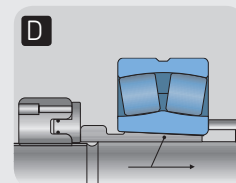
One sliding surface



One sliding surface



Two sliding surfaces



Two sliding surfaces

Patent protected

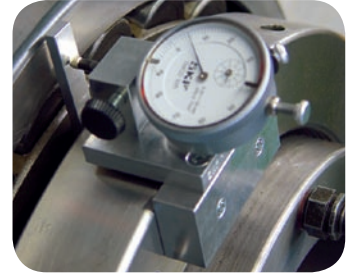
Hydraulic Nut Drive-up Adapter HMVA 42/200

For use with previous generation of SKF HMV(C) hydraulic nuts

The SKF Drive-up Method is the preferred method for mounting SKF Spherical Roller and CARB Bearings on tapered seatings. In conjunction with an SKF dial indicator, the adapter allows the previous generation of HMV nuts to be used with the SKF Drive-up Method.

- One adapter suits previous generation nuts from HMV(C) 42 up to 200
- Rugged construction
- Easy to attach to the HMV nut using strong magnets
- Used in conjunction with SKF dial indicators

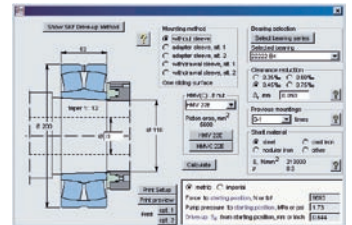
The adapter can be used with nuts from size HMV(C) 42 to HMV(C) 200. The adapter is not required for the current generation of HMV(C) ..E nuts.



SKF Drive-up Method CD-ROM

A computerised handbook on mounting bearings with a tapered bore

The SKF Drive-up Method is used for mounting bearings with a tapered bore. This CD-ROM gives a description of the method with the aid of pictures, animations and tables. The program includes calculation modules covering most bearing mounting situations in seven languages.



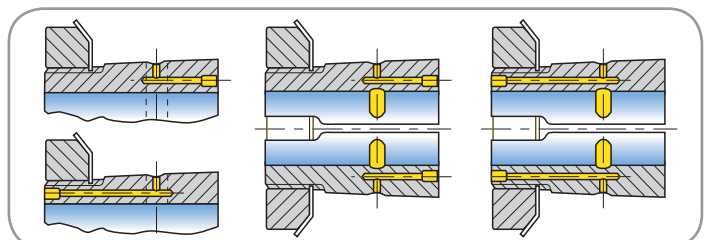
Adapter and withdrawal sleeves for oil injection

Mounting bearings made easy

These SKF sleeves facilitate the use of the SKF Oil Injection Method. The larger sleeves have oil supply ducts and distribution grooves, enabling the user to inject oil between the sleeve and bearing bore and between the sleeve and the shaft. This oil reduces friction and force necessary for mounting, particularly when mounting in a dry state.

- Reduces the risk of damage to shaft and sleeve
- Reduces time to mount and dismount bearings
- A full range of pumps, nipples and pipes are available
- SKF sleeves also help making bearing dismounting easier

For more information, please refer to the SKF General Catalogue, the SKF Maintenance Handbook or consult an SKF application engineer.





Mounting and lubrication



Hydraulic Nuts HMV ..E series

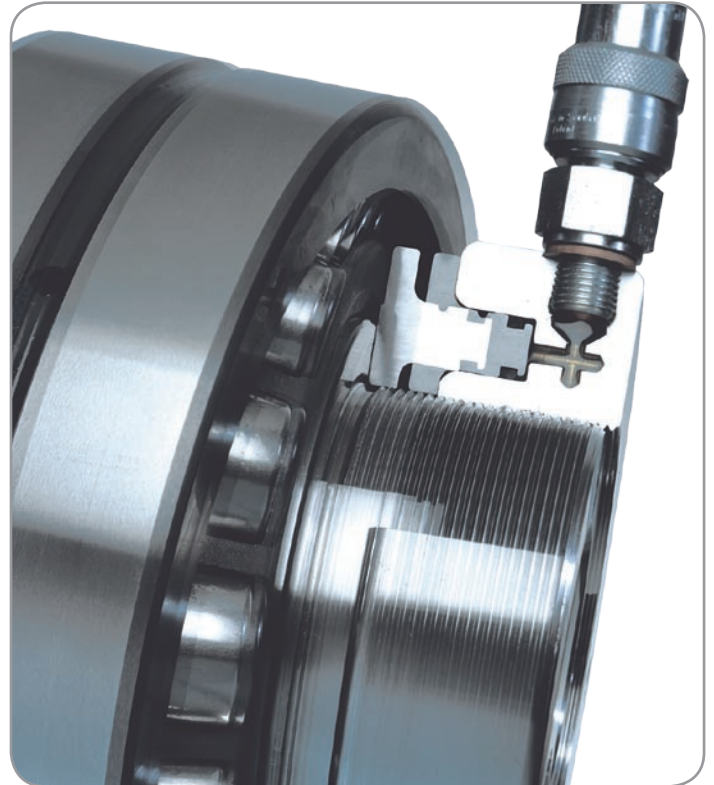
Easy application of high drive-up forces

Mounting bearings on tapered seatings can be a difficult and time-consuming job. Using an SKF hydraulic nut facilitates easy and quick application of the high drive-up forces required for mounting bearings.

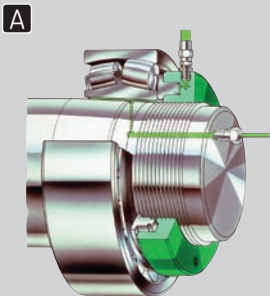
- Wide size range, covering shaft diameters from 50 to 1 000 mm as standard
- Full range of inch threads available, series HMVC ..E - 1,967 up to 37,410 in
- Quick connection coupling can be fitted on the face or side of the nut, allowing the nut to be used in areas where space is limited
- Spare set of piston seals and Maintenance kit supplied as standard
- To assist nut threading, a tube of lubricant is supplied standard with all nuts from size HMV(C) 54E
- To facilitate easy nut threading, all nuts from size HMV(C) 54E are equipped with two tommy bars and four mating holes on their front face
- Nuts from size HMV(C) 94E are equipped with eyebolts, allowing easy handling
- Nuts from size HMV(C) 94E have the starting position of the thread indicated, facilitating easy matching of thread positions on both the nut and mating thread
- Special threads and sizes available on request

All HMV ..E nuts are equipped with a quick connection coupling to fit the SKF hydraulic pumps.

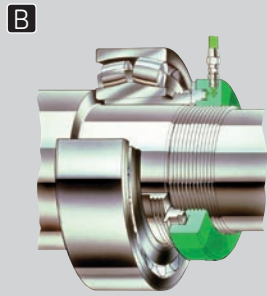
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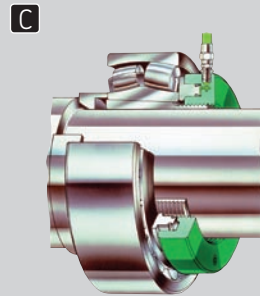
In addition to facilitating the mounting of bearings, SKF hydraulic nuts HMV E series can be also used to assist during dismantling them. See page 119 of this catalogue for more details.



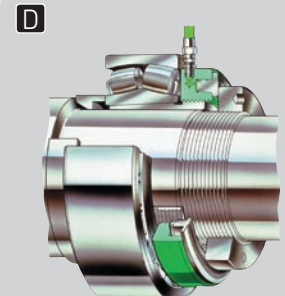
A HMV ..E nut for driving the bearing onto a tapered seating.



B HMV ..E nut screwed onto the shaft for driving in a withdrawal sleeve.



C HMV ..E nut for driving the bearing onto an adapter sleeve.



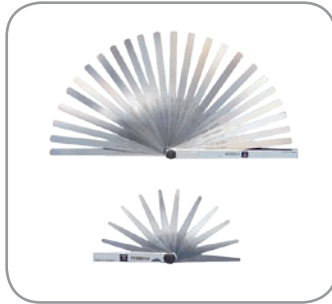
D HMV ..E nut and special stop nut for driving in a withdrawal sleeve.

Feeler Gauges 729865 series

For accurate bearing clearance measurement

SKF Feeler Gauges are designed to measure the internal clearance when adjusting spherical roller bearings. Two types are available, one with 13 blades of 100 mm (4 in) length and the other with 29 blades of 200 mm (8 in) length.

- High accuracy of measurement
- 729865 A is supplied with protective plastic cover
- 729865 B is supplied with protective steel cage



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SensorMount Indicator TMEM 1500

The tool to monitor the mounting of SensorMount bearings

The SensorMount Indicator TMEM 1500 provides a direct reading of the fit of a “SensorMount” bearing mounted on a tapered seating. The SensorMount Indicator is only compatible with SKF bearings, which are fitted with the SensorMount sensor. These bearings from SKF have the designation suffixes ZE, ZEB, or ZEV, e.g. ZE 241/500 ECAK30/W33. The SensorMount Indicator provides a numeric value, which guides the user in achieving a reliable bearing fit.

SKF bearings fitted with the SensorMount system can also be mounted on adapter sleeves, withdrawal sleeves and hollow shafts. The material composition of the shaft has no effect on the proper operation of the SensorMount system.

What you see is what you get; directly indicates the real reduction in internal bearing clearance

- Easy to use
- Fast
- Reliable
- Simplifies the mounting process:
 - No calculations needed
 - Makes feeler gauges obsolete
 - Minimises the risk of human errors



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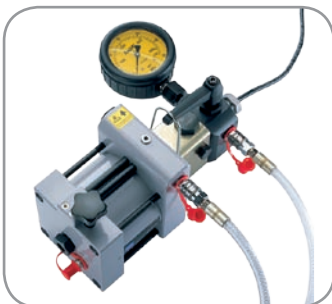


Mounting and lubrication

Hydraulic pumps and oil injectors selection guide

Max. working pressure	Pump	Type	Oil container capacity	Connection nipple	Mounting Applications*
30 MPa 4 350 psi	THAP 030	Air-driven pump	Separate container oil	G 3/4	OK Couplings
50 MPa 7 250 psi	TMJL 50	Hand operated pump	2 700 cm ³ (165 in ³)	G 1/4	All HMV ..E nuts OK Couplings
100 MPa 14 500 psi	729124	Hand operated pump	250 cm ³ (15 in ³)	G 1/4	≤ HMV 54E
	TMJL 100	Hand operated pump	800 cm ³ (48 in ³)	G 1/4	≤ HMV 92E
150 MPa 21 750 psi	THAP 150	Air-driven pump	Separate container	G 3/4	Bolt tensioners, propeller mounting
	728619 E	Hand operated pump	2 550 cm ³ (155 in ³)	G 1/4	All HMV ..E nuts
300 MPa 43 500 psi	THAP 300E	Air-driven pump	Separate container	G 3/4	OK Couplings Large pressure joints
	226400	Hand operated oil injector	200 cm ³ (12,2 in ³)	G 3/4	OK Couplings Adapter / withdrawal sleeves
	729101 B	Oil injection kit	200 cm ³ (12,2 in ³)	Several	Many applications, such as OK Couplings Adapter / withdrawal sleeves
	TMJE 300	Oil injection set	200 cm ³ (12,2 in ³)	Several	
	226270	Screw injector	5,5 cm ³ (0,33 in ³)	G 3/8	Machine tool applications shaft diameter ≤ 100 mm
	226271	Screw injector	25 cm ³ (1,5 in ³)	G 3/4	Machine tool applications shaft diameter ≤ 200 mm
400 MPa 58 000 psi	THAP 400E	Air-driven pump	Separate container	G 3/4	OK Couplings Large pressure joints
	226400/ 400 MPa	Hand operated oil injector	200 cm ³ (12,2 in ³)	G 3/4	Joints with high interference fits
	729101 E	Oil injection kit	200 cm ³ (12,2 in ³)	G 1/4	Complete kit / set to suit many applications
	TMJE 400	Oil injection set	200 cm ³ (12,2 in ³)	G 1/4	Complete kit / set to suit many applications

*The mounting applications given above are for guidance only. The interference fit present may mean that a pump / injector with a higher-pressure capacity is required.



Hydraulic Pump 729124

100 MPa (14 500 psi)



The 729124 pump is suitable for use with hydraulic nuts ($\leq$ HMV 54E) to mount bearings or components where a maximum pressure of 100 MPa (14 500 psi) is required. The pump is supplied with 1 500 mm (59 in) pressure hose, quick connect coupling and mating nipple and pressure gauge.

- Suitable for hydraulic nuts $\leq$ HMV 54E
- Extra litre of mounting fluid
- Special pump configurations available
- Packed in a sturdy protective case
- Oil container capacity 250 cm³ (15 in³)

Applications

- SKF hydraulic nuts $\leq$ HMV 54E
- All other oil injection applications where the maximum pressure is 100 MPa (14 500 psi)

The pump is filled with SKF Mounting Fluid LHM 300 and is supplied with an extra litre of fluid. For applications where space does not permit the use of a quick connect coupling and nipple, such as AOH sleeves, a special pump design is available (729124 A).



Hydraulic Pump TMJL 100

Large oil container 100 MPa (14 500 psi)



The TMJL 100 pump is suitable for use with hydraulic nuts ($\leq$ HMV 92E) to mount bearings or components where a maximum pressure of 100 MPa (14 500 psi) is required. The pump is supplied with 3 000 mm (118 in) pressure hose,

- Suitable for hydraulic nuts $\leq$ HMV 92E
- Suitable with SKF hydraulic assisted pullers TMHP series
- Extra litre of mounting fluid
- Packed in a sturdy protective case
- Oil container capacity 800 cm³ (48 in³)

Applications

- SKF hydraulic nuts $\leq$ HMV 92E
- All other oil injection applications where the maximum pressure is 100 MPa (14 500 psi)
- SKF hydraulic assisted pullers TMHP series

quick connect coupling and mating nipple, and pressure gauge. The pump is filled with SKF Mounting Fluid LHM 300 and is supplied with an extra litre of fluid.





Mounting and lubrication



Hydraulic Pump TMJL 50

50 MPa (7 250 psi)

133

The TMJL 50 pump is mainly intended for use on the low-pressure side of SKF OK Couplings, but is also suitable for applications where a maximum pressure of 50 MPa (7 250 psi) is required. The pump is supplied with a 3 000 mm (118 in) long high-pressure hose with quick connect coupling

and mating nipple. It is filled with SKF Mounting Fluid LHMf 300 and is supplied with an extra litre of fluid. The pump is fitted with an over-pressure valve and has a connection port for a pressure gauge.

- Large oil container (cap. 2 700 cm³, 165 in³)
- Over pressure valve
- Extra litre of mounting fluid
- Packed in a sturdy protective case

Applications

- Low pressure side of SKF OK Couplings
- All sizes HMV ..E hydraulic nuts
- All other oil injection applications where the maximum pressure is 50 MPa (7 250 psi)



Hydraulic Pump 728619 E

150 MPa (21 750 psi)

134

The 728619 E is a two-stage pump suitable for use with SKF Supergrip bolts and to mount bearings or components where a maximum pressure of 150 MPa (21 750 psi) is required. The pump is supplied with a 3 000 mm (118 in) pressure hose, quick connect coupling and mating nipple and pressure gauge.

The pump is filled with SKF Mounting Fluid LHMf 300, is supplied with an extra litre and is protected in a sturdy metal case.

- Two stage pressure pumping
- Extra litre of mounting fluid
- Packed in a sturdy protective metal case
- Oil container capacity 2 550 cm³ (155 in³)

Applications

- SKF Supergrip bolts
- All other oil injection applications where the maximum pressure is 150 MPa (21 750 psi)
- All sizes HMV ..E hydraulic nuts



Air-driven Hydraulic Pumps, THAP series

30, 150, 300 and 400 MPa (4 350, 21 750, 43 500 and 58 000 psi)

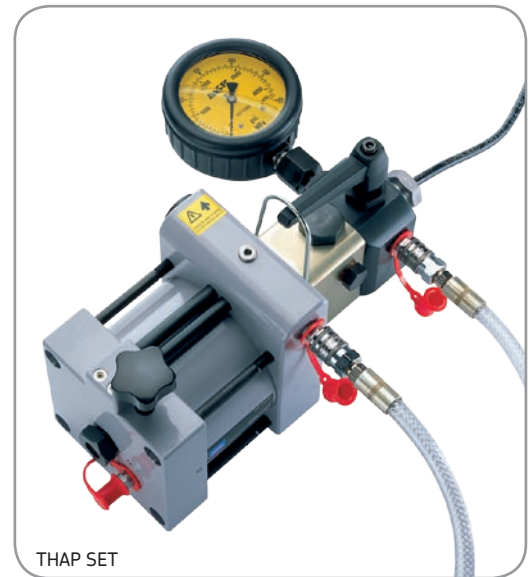
The THAP Air-driven Hydraulic Pumps are available in four different pressure versions. They can be used for mounting OK Couplings, large pressure joints such as bearings, flywheels, couplings and railway wheels. The pumps consist of a high-pressure hydraulic pump, driven by an air piston.

The units are supplied in a sturdy case including oil suction and return hoses with quick connect couplings. The pumps can also be supplied in complete sets, which consists of pump, pressure gauge, adapter block, high-pressure pipe and connection nipples.

- Time savings compared to hand operated pumps
- Portable
- Continuous supply of oil
- Sturdy storage boxes
- Low, medium and high pressure units

Applications

- SKF OK Couplings
- Mounting bearings
- Mounting ship propellers, rudder pintles, railway wheels and other similar applications



Screw Injectors 226270 and 226271

300 MPa (43 500 psi)

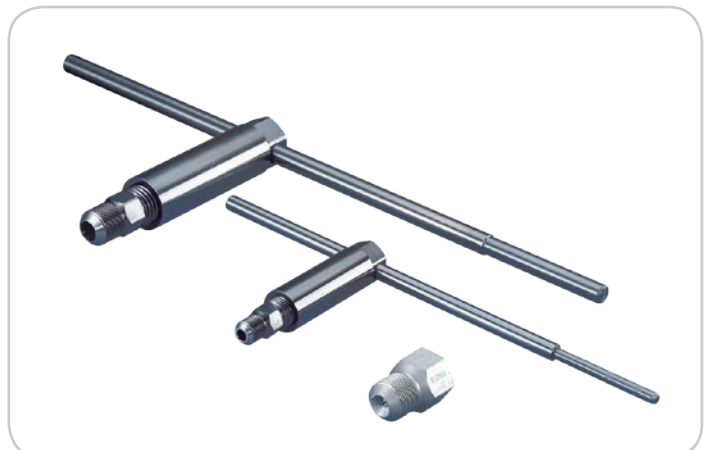
The SKF Screw Injectors 226270 and 226271 are mainly used within the machine tool industry for mounting bearings and components using the SKF Oil Injection Method. Valve nipples 226272 and 226273 can be used to retain the oil pressure while the injector is refilled.

226270

- Suitable for components with a shaft diameter up to 100 mm (4 in)
- Oil container capacity 5,5 cm³ (0,33 in³)

226271

- Suitable for components with a shaft diameter not exceeding 200 mm (8 in)
- Oil container capacity 25 cm³ (1,5 in³)





Mounting and lubrication



Oil Injector 226400 series

300 and 400 MPa (43 500 and 58 000 psi)

The 226400 series oil injectors have a varied usage when applying the SKF Oil Injection Method. For mounting bearings, couplings, railway wheels, gear wheels, flywheels, ship propellers and so on. The injector is supplied with an oil reservoir in a compact carrying case.

- Easy to operate
- Compact carrying case
- Large range of accessories available, including:
 - Adapter block
 - Pressure gauges
 - High pressure pipes
 - Connecting nipples
- Oil container capacity 200 cm³ (12,2 in³)

Applications

- For mounting and dismantling of:
 - Bearings
 - Couplings
 - Railway wheels
 - Gear wheels
 - Flywheels
 - Ship propellers and so on
- For any oil injection application where maximum pressure of up to 400 MPa (58 000 psi) is required

For applications where 400 MPa (58 000 psi) is required a special model is available: 226400/400 MPa. The injector can be mounted directly onto the work piece or connected to an adapter block to make it a floor standing model making it easy to connect pressure gauges and high-pressure pipes.



Oil Injection Kits 729101 series

300 and 400 MPa (43 500 and 58 000 psi)

The oil injection kits contain the oil injector 226400 complete with high pressure pipe, pressure gauge, adapter block and a range of connection nipples all packed together in a sturdy plastic carrying case.

- Complete high-pressure kits, including oil injector, pressure gauge, 2,0 m high-pressure pipe and a range of connection nipples
- Oil container capacity 200 cm³ (12,2 in³)



Ordering details

Designation	Description
729101 B	Oil injection kit (300 MPa / 43 500 psi)
729101 E	Oil injection kit (400 MPa / 58 000 psi)

Contents list

Designation	729101 B	729101 E
Oil injector	226400	226400/400 MPa
Adapter block	226402	226402
High pressure pipe (G 3/4 - 1/4)	227957 A	227957 A/400 MPa
Connection nipple (G 1/4 - 1/8)	1014357 A	-
Connection nipple (G 1/4 - 1/2)	1016402E	1016402E
Connection nipple (G 1/4 - 3/4)	228027E	228027E
Pressure gauge (0 - 300 MPa)	1077589	1077589/2 (0-400 MPa)
Carrying case	729111 B	729111 B

Oil Injection Sets TMJE 300 and 400 series

300 and 400 MPa (43 500 and 58 000 psi)

The SKF TMJE 300 and 400 are used for mounting of pressure joints of all sizes and applications such as propellers, rolling bearings, couplings, gears, pulleys, flywheels and SKF OK couplings.

- Complete high-pressure set with integral pressure gauge, oil reservoir and 2,0 m high-pressure pipe
- Can be dismantled and used directly on the application
- Provided with range of connection nipples
- Oil container capacity 200 cm³ (12,2 in³)



Contents list

Designation	TMJE 300	TMJE 400
Oil injector	TMJE 300-1	TMJE 400-1
Pressure gauge	1077589	1077589/2
High pressure pipe (G 3/4 - 1/4)	227957 A	227957 A/400MPa
Connection nipple (G 1/4 - 1/8)	1014357 A	-
Connection nipple (G 1/4 - 1/2)	1016402E	1016402E
Connection nipple (G 1/4 - 3/4)	228027E	228027E
Carrying case	728245/3A	728245/3A
Plug	729944E	729944E
Mounting fluid	LHMF 300/1	LHMF 300/1

Hydraulic accessories

Adapter Block 226402

The adapter block 226402 consists of a cast steel block to which a pressure gauge and high-pressure pipe can be connected. It comes with a floor support and a 90 degree connection nipple for the oil reservoir.





Mounting and lubrication



High-pressure Pipes

Maximum working pressure 300 MPa (43 500 psi)

The range of SKF High-pressure Pipes covers most applications where there is a requirement to transfer oil at high pressure. They consist of a steel pipe with a steel ball fitted to both ends.

- Wide range of pipes
- All pipes are pressure tested to 100 MPa (14 500 psi) over recommended working pressure
- Special lengths (up to 4 m / 157 in) made on request
- 400 MPa versions available

Safety note:

For safety reasons, these high-pressure pipes have a maximum recommended service life. All SKF high pressure pipes are hard-marked with the year in which their recommended service life expires; e.g. RECOMMENDED SERVICE LIFE EXPIRES 2015.

Two swivelling connection nipples press these balls against the seating of the connecting hole and thus sealing against oil leakage.



Pressures Gauges

100 to 400 MPa (14 500 to 58 000 psi)

SKF Pressure Gauges are designed to fit SKF hydraulic pumps and oil injectors. The gauges are all liquid filled and/or equipped with a restriction screw in order to absorb any sudden pressure drop and prevent damage.

- Covers pressures of 100 to 400 MPa (14 500 to 58 000 psi)
- Protection against sudden pressure drops
- Safety glass and blow out discs on all gauges
- Stainless steel case
- Dual scales MPa/psi
- Easy to read, high visibility yellow gauge faces

Safety glass and blowout discs are standard for all gauges and all have dual scales (MPa/psi).



Plugs for oil ducts and vent holes

Up to 400 MPa (58 000 psi)

SKF plugs have been designed to seal off oil connections at a maximum pressure of 400 MPa (58 000 psi).



Flexible High-pressure Hoses

Maximum working pressure up to 150 MPa (21 750 psi)

The SKF flexible pressure hoses are designed to be used together with the quick connect coupling 729831 A and nipple 729832 A on the range of SKF hydraulic pumps.

Safety note:

All flexible pressure hoses are subject to ageing and after a number of years the performance deteriorates. All SKF flexible pressure hoses are hard marked with the year in which their life expires, e.g. LIFE EXPIRES 2015.



Quick connecting coupling and nipples

For easy pressure hose connection

One coupling and two different nipples are available to connect SKF hydraulic pumps to the work piece. When nipples with other thread types are required, select an additional SKF nipple from the range to make the connection. Nipple 729832 A is supplied standard with all SKF HMV ..E hydraulic nuts.



Connection nipples with metric and G pipe threads

External to internal metric and G pipe threads

SKF provides a wide range of connecting nipples covering many different thread combinations and sizes. All nipples with an E suffix have a maximum working pressure of 400 MPa (58 000 psi). The remainder have a maximum working pressure of 300 MPa (43 500 psi).

Connection nipples with NPT tapered threads

Connection nipples with tapered threads (NPT) and pipe threads (G)

SKF can also supply a range of adapters for connecting NPT threads to G threads. All nipples have a maximum working pressure of 300 MPa (43 500 psi). Nipples having a maximum working pressure of 400 MPa (58 000 psi) are available on request.





Mounting and lubrication



Extension pipes with connecting nipples

Catering for difficult connection applications



M4 extension pipe with connection nipple

Used to extend a high-pressure pipe with a G 1/4 nipple (e.g. 227957 A) when the connection hole has an M4 thread. The extension pipe and connection nipple should be ordered as separate items.

M6 extension pipe with connection nipple

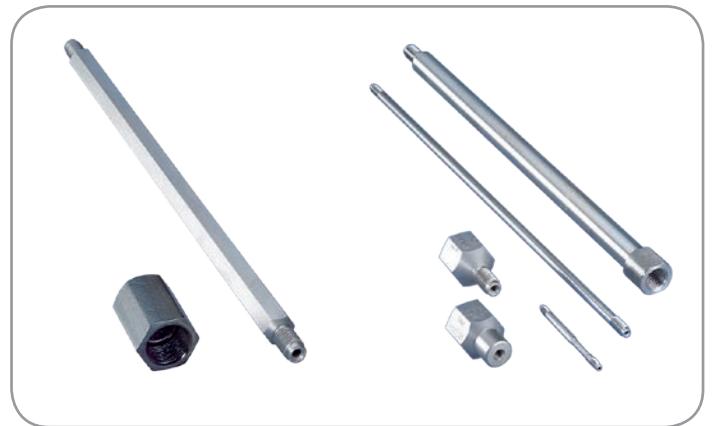
Used to extend a high-pressure pipe with a G 1/4 nipple (e.g. 227957 A) when the connection hole has a M6 thread. The extension pipe and connection nipple should be ordered as separate items.

Valve nipple with extension pipe

This combination is intended to be used between an oil pressure joint and an oil injector (226271) when a thin wall thickness of the pressure joint prevents the connection of the injector directly to the joint. The valve nipple is used to retain the pressured oil while the injector is refilled. The extension pipe and connection nipple should be ordered as separate items.

Extension pipe

This unit is used for connection to components with a thin wall thickness, such as sleeves with oil injection preparations. It is normally used in combination with high-pressure pipes such as 227957 A.



Mounting Fluid LHM 300

For easy and quick bearing mounting

The SKF Mounting Fluid LHM 300 is suitable for use with SKF hydraulic equipment, including hydraulic pumps, HMV ..E nuts and oil injection tools.

The LHM 300 contains anti corrosives which are non aggressive to seal materials such as nitrile rubber, perbunan, leather and chrome leather, PTFE, and so on.

Ordering details and technical data

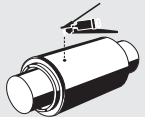
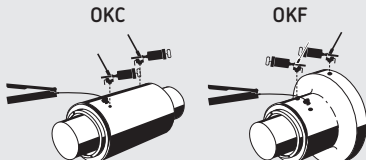
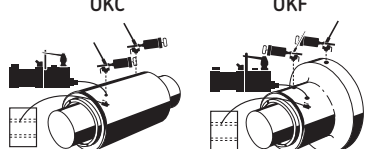
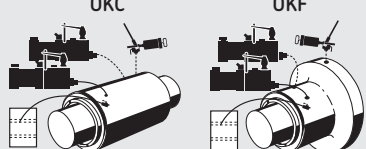
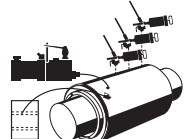
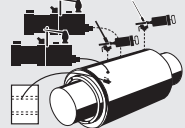
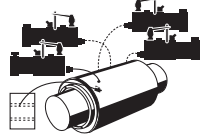
Designation	LHM 300/pack size
Specific gravity	0,882
Flash point	200 °C (390 °F)
Pour point	-30 °C (-22 °F)
Viscosity at 20 °C (68 °F)	300 mm ² /s
Viscosity at 40 °C (104 °F)	116 mm ² /s
Viscosity at 100 °C (212 °F)	17,5 mm ² /s
Viscosity index	160
Available pack size	1, 5, 205 litre



OK Coupling mounting and dismounting kits

Specially prepared kits are made to simplify the process of mounting and dismounting SKF OK Couplings.

Ordering details and selection chart

Coupling size	Designation	Contents	Weight	Application
OKC 25 - OKC 90	TMHK 35	1 × TMJE 300-1 Oil injector set 1 × 729944 E Plug 1 × 227958A Pressure pipe (for OKC 80 and 90) 1 × 729123A/2000 Pressure pipe (for OKC 25 - 75) Tools and storage case	13,8 kg (30,4 lb)	
OKC 100 - OKC 170 OKCS 178 - OKCS 360	TMHK 36	1 × 226400 Injector with spares 1 × TMJL 50 Hydraulic pump Tools and storage case	19 kg (41,8 lb)	
OKC 180 - OKC 250 OKF 100 - OKF 300 * = for use with OKF couplings	TMHK 37	2 × 226400 Injector with spares 1 × 226402 * Adapter block 1 × 227958A * High pressure pipe 1 × TMJL 50 Hydraulic pump Tools and storage case	28,1 kg (61,8 lb)	
OKC 180 - OKC 490 OKF 300 - OKF 700 Shipboard or infrequent use	TMHK 38	1 × THAP 030/SET Air-driven pump set 1 × 729147A Return hose 2 × 226400 Injector with spares	32,1 kg (70,6 lb)	
OKC 180 - OKC 490 OKF 300 - OKF 700 Shipyard or frequent use	TMHK 38S	1 × THAP 030/SET Air-driven pump set 1 × 729147A Return hose 1 × THAP 300E Air-driven pump 1 × 226400 Injector with spares	78,2 kg (172,3 lb)	
OKC 500 - OKC 600 Shipboard or infrequent use	TMHK 39	1 × THAP 030/SET Air-driven pump 1 × 729147A Return hose 3 × 226400 Injector with spares	35,1 kg (77,2 lb)	
OKC 500 and larger Shipboard or infrequent use	TMHK 40	1 × THAP 030/SET Air-driven pump 1 × THAP 300E Air-driven pump 1 × 729147A Return hose 2 × 226400 Injector with spares	80,2 kg (176,7 lb)	
OKC 500 and larger Shipyard or frequent use	TMHK 41	1 × THAP 030/SET Air-driven pump 3 × THAP 300E Air-driven pump 1 × 729147A Return hose	132,7 kg (293,3 lb)	



Accessories

Anti Corrosive Agent LHRP 1

SKF LHRP 1 provides excellent long-term corrosion protection to ferrous and non-ferrous metals. When applied, it leaves a stable rust protection film over the metal component.

- Excellent rust protection in high humidity environments (tests at 30 °C/80 °F – 90 % relative humidity indicates full protection for at least one year)
- Excellent long-term indoor storage protection



Special Working Gloves TMBA G11W

For providing protection while maintaining excellent grip

The SKF Special Working Gloves TMBA G11W are specially designed for general-purpose industrial maintenance work. The inside palm of the gloves are coated with non-flammable dots

providing excellent grip.

- Abrasion resistant
- Blade cut resistant
- Tear resistant
- Puncture resistant
- Flexible and comfortable gloves with excellent grip
- Lint free
- Tested and certified according to EN 388 (mechanical risks)



Heat Resistant Gloves TMBA G11

For safe handling of heated components up to 150 °C (302 °F)

The SKF Heat Resistant Gloves TMBA G11 are specially designed for the handling of heated bearings. They are made of special fabric to obtain the following combination of features:

- Lint free
- Heat resistant up to 150 °C (302 °F)
- Cut resistant
- Contains no asbestos
- Tested and certified for mechanical risks (EN 388) and thermal risks (EN 407)



Extreme Temperature Gloves TMBA G11ET

For safe handling of heated components up to 500 °C (932 °F)

The TMBA G11ET gloves are especially designed for allowing the safe handling of heated bearings or other components for prolonged periods. They can withstand extreme temperatures of up to 500 °C (932 °F), without the presence of hot liquid or steam, with a high degree of non-flammability.

- Heat-resistance to extreme temperatures allows the safe and prolonged handling heated components
- High-degree of non-flammability reduces the risk of burning
- Extremely tough KEVLAR gloves with high cut, abrasion, puncture and tear resistance for increased safety
- Lint-free design safeguards against bearing contamination
- Comfortable to wear, as they are knitted from flexible high performance materials in one piece without seams
- Tested and certified for mechanical risks (EN 388) and thermal risks (EN 407)



Heat and Oil Resistant Gloves TMBA G11H

For safe handling of oily and heated components up to 250 °C (482 °F)

The SKF Heat and Oil Resistant Gloves TMBA G11H are specially designed for the handling of hot and oily bearings. They are made of multiple layers of different kinds of fabric to obtain an important combination of features:

- A combination of heat, cut, oil and water resistance
- KEVLAR gloves
- Melt and burn proof
- Maximum temperature: 250 °C (482 °F)
- Suitable for submerging in liquids of a temperature up to 120 °C (248 °F) (e.g. hot oil bath)
- Remains heat resistant when wet
- Cut resistant
- Lint free
- Tested and certified for chanical risks (EN 388) and thermal risks (EN 407)





Mounting and lubrication

Lubrication

SKF greases for most bearing applications

Sealed bearings are pre-lubricated and do not require lubrication when mounted. However, in applications where open bearings are used, these bearings must be lubricated after mounting. Selecting the right bearing grease

for your application is another step to maximise your bearing's service life. SKF offers 15 different bearing greases, which have been especially developed by SKF to cover most bearing applications.

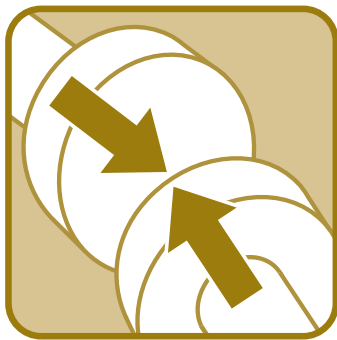
SKF bearing greases

Designation	Description
LGMT 2	All purpose industrial and automotive bearing grease
LGMT 3	All purpose industrial and automotive bearing grease
LGEP 2	Extreme pressure bearing grease
LGFP 2	Food compatible bearing grease
LGFB 2	Food compatible, high load bearing grease
LGEM 2	High viscosity bearing grease with solid lubricant
LGEV 2	Extremely high viscosity bearing grease with solid lubricant
LGLT 2	Low temperature, extremely high speed bearing grease
LGFL 1	Food compatible, low temperature bearing grease
LGGB 2	Green biodegradable grease
LGWM 1	Extreme pressure low temperature bearing grease
LGWA 2	Wide temperature range bearing grease
LGHB 2	High viscosity, high temperature bearing grease
LGHP 2	High performance bearing grease
LGET 2	Extreme temperature bearing grease



For more information about the above listed SKF bearing greases, please see the Re-lubrication section of this catalogue on pages 49 - 82





Alignment

Misalignment costs time and money	42
Shaft Alignment Tools TMEA series	44
Shaft Alignment Tool TMEA 2	44
Shaft Alignment Tool with printer capability TMEA 1P/2.5	45
Intrinsically Safe Shaft Alignment Tool TMEA 1PEX	45
Thermal Printer TMEA P1	46
Machinery Shims TMAS series	46
Belt Alignment Tool TMEB 2	48



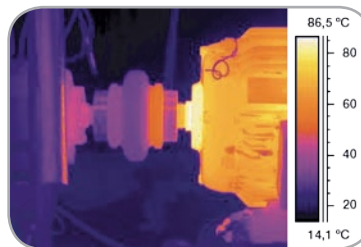
Alignment



Misalignment costs time and money

Shaft misalignment

Shaft misalignment is responsible for up to 50% of all costs related to rotating machinery breakdowns. These breakdowns increase unplanned machinery downtime, resulting in higher maintenance costs and loss of production. Additionally, misaligned shafts can increase vibration levels and friction, which can significantly increase energy consumption and can cause premature bearing failures.



Overheated motor due to misalignment.
Picture taken using a FLIR infrared camera.

Traditional shaft alignment methods

Traditional alignment methods, although very common, do not often produce the exacting degree of accuracy required by today's precision machinery. The rough alignment methods still used nowadays, such as using a straight edge and feeler gauge, may be quick, but they can be inaccurate.

Another traditional method employing dial indicators offers a higher degree of accuracy, but it requires specialist operators and can be time consuming.



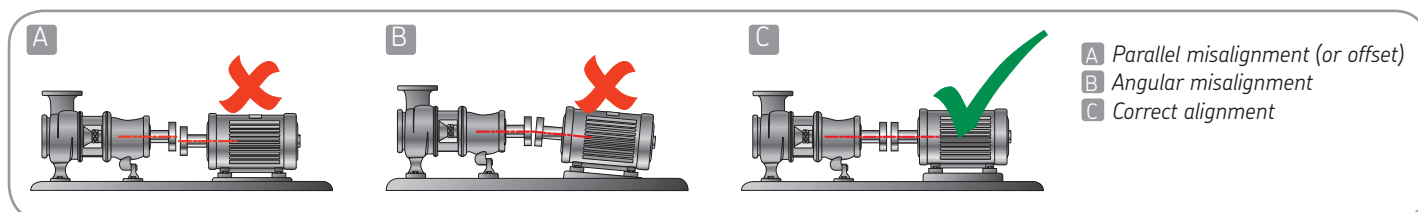
Laser shaft alignment method

Laser alignment methods are a marked improvement on traditional ones. A laser shaft alignment tool facilitates a quicker and more accurate alignment than traditional methods. Since shaft misalignment has a direct, negative, impact on bearing service life, SKF offers a range of high precision,

easy-to-use laser shaft alignment tools. The tools, the TMEA series, combine simplicity with a high degree of accuracy. They feature a three-step process for correcting alignment: measuring, aligning and documenting.

Accurate shaft alignment can help you:

- Increase bearing life
- Reduce stress on couplings and thereby the risk of overheating and breakage
- Reduce wear on seals, helping to prevent contamination and lubricant leakage
- Reduce friction and thereby energy consumption
- Reduce noise and vibration
- Increase machinery uptime, efficiency and productivity
- Reduce costs of replacing components and machinery downtime

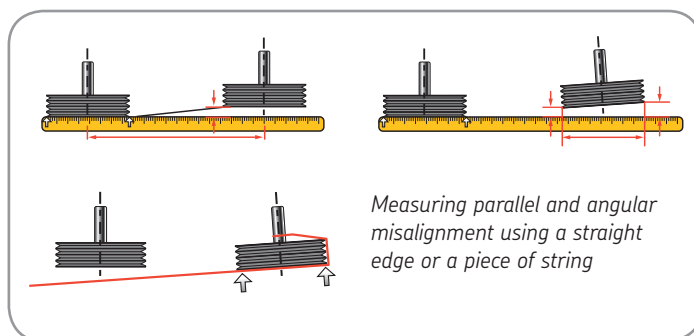


Belt misalignment

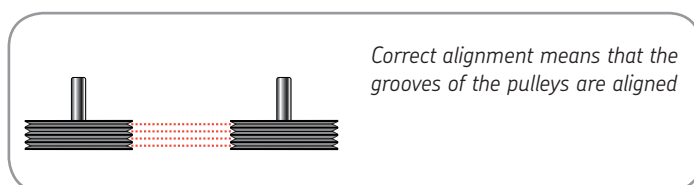
One of the common reasons for unplanned downtime of belt-driven machinery is pulley misalignment. Pulley misalignment can increase wear on pulleys and belts as well as increasing the noise and vibration levels, which can result in unplanned machinery downtime. Another side effect of increased vibration is premature bearing failure. This too can cause unplanned machinery downtime.

Traditional belt alignment methods

These methods, which are the most widely used, involve either using visual judgement alone or visual judgement in combination with a straight edge and/or length of string. The advantage of these traditional methods is the perceived short time needed for adjustment, although the use of a straight edge takes more time than visual judgement alone. The major disadvantage is the lack of accuracy.



Some belt manufacturers recommend a maximum horizontal angle misalignment of $0,5^\circ$ or even $0,25^\circ$, which is impossible to accomplish by using naked eye.



Laser belt alignment methods

A laser belt alignment tool facilitates alignment with far more speed and accuracy than with traditional methods. Belt alignment tools available on the market can be categorised according to the way the tools can be attached to the pulley and the way they align. In general there are two groups; one aligns the face of the pulleys and the other aligns the grooves of the pulleys.

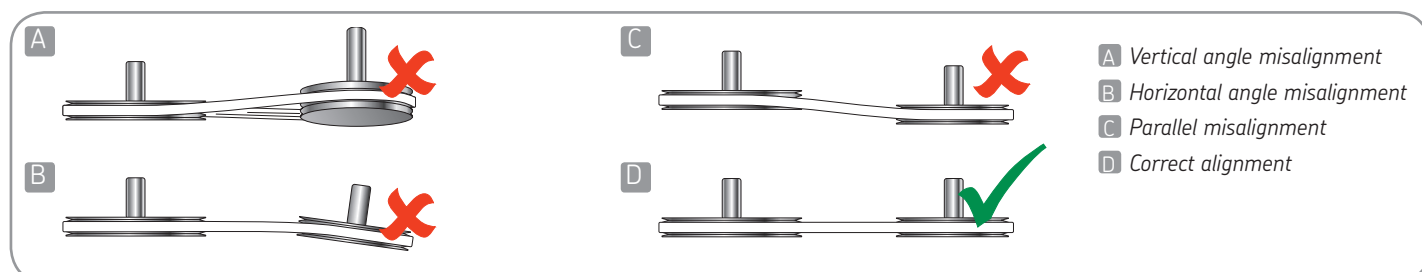
The major disadvantage of the tools, which use the face or side of the pulley as a reference for aligning the pulleys and belts, is that only the face of the pulley is used as a reference.

This means that only the faces of the pulleys are aligned with each other and not necessarily the grooves in which the belt runs. This method results in varying degrees of accuracy when the pulleys are of different thickness, brands or kinds.

The tools, which align the grooves of the pulley allow for alignment where it counts most – in the grooves of the pulley, substantially increasing accuracy irrespective of the thickness, brand or type of pulleys.

Accurate pulley and belt alignment can help you:

- Increase bearing life
- Increase machinery uptime, efficiency and productivity
- Reduce wear on pulleys and belts
- Reduce friction and thereby energy consumption
- Reduce noise and vibration
- Reduce costs of replacing components and machinery downtime





Alignment



Shaft Alignment Tools TMEA series

Pinpoint accurate alignment simply achieved

The SKF Shaft Alignment Tools TMEA Series offer you simplicity with a high degree of accuracy. These highly innovative tools feature a three-step process for correcting alignment: measuring, aligning and documenting. First, measure the machinery's current alignment status.

- Easy-to-use, three-step process: measure – align – document
- Compact, lightweight design
- Spirit levels allow easy and fast positioning of the measuring units
- Selectable mm or inch reading of measurement facilitates worldwide use
- Supplied in sturdy, lightweight carrying cases for portability
- Supplied with high precision SKF pre-cut shims for accurate alignment

Then align the machine vertically and horizontally. Finally, document and keep track of the alignment activities. These three simple steps allow you to easily and effectively align shafts using laser technology.



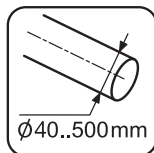
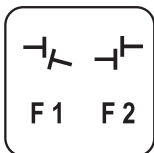
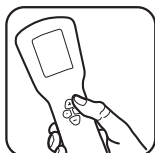
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Shaft Alignment Tool TMEA 2

Easy, quick and affordable shaft alignment

The TMEA 2 is an easy-to-use shaft alignment tool, which requires no special training to operate. The two measuring units can be easily attached to the shafts using magnetic brackets or chains. Each measuring unit emits a laser line, which is projected on the detector of the other unit.

- Display unit simultaneously provides clear “real-time” coupling and feet values during alignment process making rechecking of the alignment unnecessary
- The laser and scale lines facilitate easy pre-alignment
- “Soft foot” feature easily guides the operator through this function
- Display unit can be held using one hand, freeing the operator to perform the alignment
- Magnetic brackets allow easy fixture of the measuring units onto the shaft
- A set of blank alignment reports to help you keep record of your alignment jobs
- Maximum distance of 0,85 m (2,8 ft) between the measuring units brackets



Shaft Alignment Tool with printer capability TMEA 1P/2.5

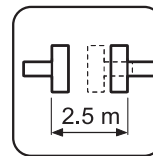
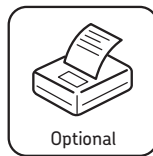
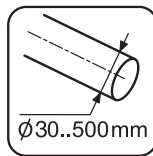
Record alignment activities using an optional printer

The TMEA 1P/2.5 offers you the advantage of keeping record of the alignment activities. It is equipped with a printer port to which the optional thermal printer TMEA P1 can be connected. The printer provides a clear and complete alignment report,

which can be used to document alignment activities.

This user-friendly printer is operated with the touch of a single button on the display unit of the TMEA 1P/2.5.

- Optional printer facilitates recording of alignment activities
- Maximum distance of 2,5 m (8,2 ft) between the measuring units makes it suitable for aligning variety of applications
- Display unit provides clear "real-time" values during the alignment process making rechecking alignment unnecessary
- User-friendly display unit with only four buttons for operation
- Supplied with blank alignment reports for recording alignment activities in case the printer is not purchased



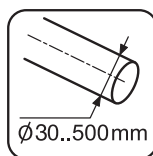
Intrinsically Safe Shaft Alignment Tool TMEA 1PEx

Accurate alignment in explosive hazardous areas

The TMEA 1PEx is an intrinsically safe (Ex) shaft alignment tool, especially designed for use in potentially explosive hazardous areas. It has been tested and certified according to the latest ATEX standards in intrinsic safety zones generally

found in industries such as the petrochemical, gas and pharmaceutical among others. The TMEA 1PEx is supplied standard with a thermal printer for recording alignment activities.

- Intrinsically safe classification ATEX code: II 2 G, EEx ib IIC T4, at ambient temperature range of 0 to 40 °C (32 to 104 °F) EC Type Examination Certificate Nemko03ATEX101X
- Standard printer facilitates recording of alignment activities
- Maximum distance of 1 m (3 ft) between the measuring units makes it suitable for aligning a variety of applications
- Display unit provides clear "real-time" values during the alignment process making rechecking alignment unnecessary
- User-friendly display unit with only five buttons for operation





Alignment



Thermal Printer TMEA P1

Keep track of alignment jobs

This compact thermal printer helps you to document your alignment jobs. A clear and complete printout of the measurement data shows that the machine has been properly aligned within the allowed tolerances.

- Compact easy-to-use printer
- Clear easy-to-read printout
- Pre-alignment and post-alignment reports possible
- Battery is rechargeable
- Continental European adaptor included
- Printer uses standard thermal paper roll (120 mm × 20 m) / (4,4 in × 65 ft)
- Can be used in combination with TMEA 1P/2.5 and TMEA 1PEx only



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Machinery Shims TMAS series

For accurate vertical machinery alignment

Accurate machine adjustment is an essential element of any alignment process. SKF single slot pre-cut shims are available in five different dimensions and in ten different thicknesses.

- Made of high quality stainless steel, allowing re-use
- Easy to fit and to remove
- Close tolerances for accurate alignment
- Thickness clearly marked on each shim
- Fully de-burred
- Pre-cut shims are supplied in packs of 10 and complete kits are also available



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Contents TMAS shim kits

TMAS 340

Thickness (mm)	0,05	0,10	0,20	0,25	0,40	0,50	0,70	1,00	2,00
Size (mm)	Quantities:								
100 × 100	20	20	20	20	20	20	20	20	10
125 × 125	20	20	20	20	20	20	20	20	10

TMAS 360

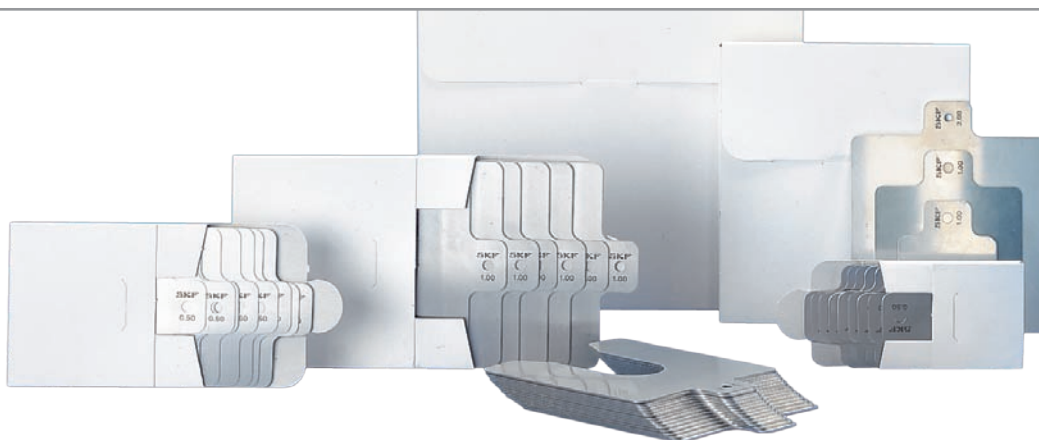
Thickness (mm)	0,05	0,10	0,25	0,50	1,00	2,00
Size (mm)	Quantities:					
50 × 50	20	20	20	20	20	20
75 × 75	20	20	20	20	20	20
100 × 100	20	20	20	20	20	20

TMAS 510

Thickness (mm)	0,05	0,10	0,20	0,25	0,40	0,50	0,70	1,00	2,00
Size (mm)	Quantities:								
50 × 50	20	20	20	20	20	20	20	20	10
75 × 75	20	20	20	20	20	20	20	20	10
100 × 100	20	20	20	20	20	20	20	20	10

TMAS 720

Thickness (mm)	0,05	0,10	0,20	0,25	0,40	0,50	0,70	1,00	2,00
Size (mm)	Quantities:								
50 × 50	20	20	20	20	20	20	20	20	20
75 × 75	20	20	20	20	20	20	20	20	20
100 × 100	20	20	20	20	20	20	20	20	20
125 × 125	20	20	20	20	20	20	20	20	20



TMAS 340



TMAS 360



TMAS 510



TMAS 720



Alignment



Belt Alignment Tool TMEB 2

Belt-driven machinery downtime caused by misalignment is a thing of the past

The SKF TMEB 2, aligns the pulleys where it counts most – in the grooves. V-guides and powerful magnets allow the TMEB 2 to be fitted in the grooves of the pulley. With only two components, a laser-emitting unit and a receiver unit, the belt alignment tool is easy and fast to attach. The three-dimensional target area on the receiver unit allows the easy

detection of misalignment as well as its nature; whether it is horizontal, vertical, parallel or a combination of all three. Armed with this precise information, the operator can easily make the appropriate adjustments until the laser line corresponds with the reference line on the receiver unit.

Versatile and user-friendly:

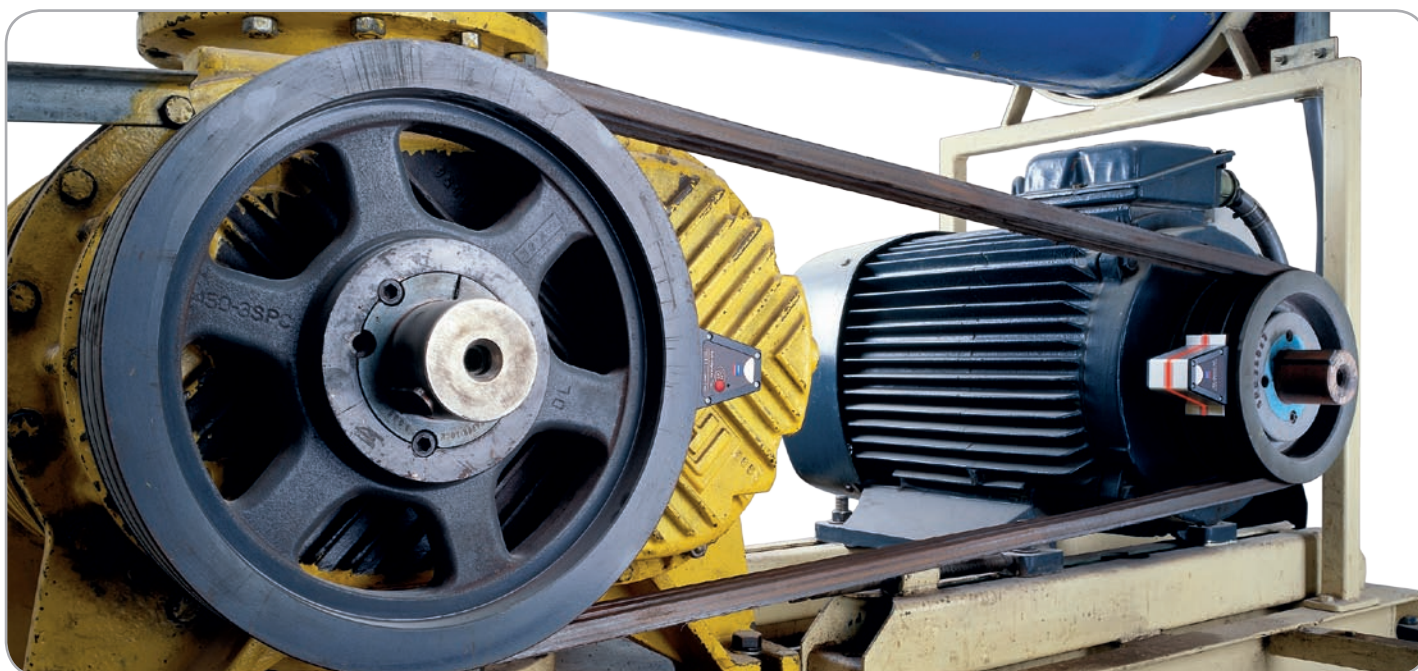
- Powerful magnets allow fast and easy attachment
- Easy-to-use, requires no special training to operate
- Three-dimensional target area simplifies the alignment process
- Facilitates simultaneous adjustment of tension and alignment
- V-guides facilitate the alignment of a wide range of V-belt pulleys
- Special side adaptor allowing alignment of multi-ribbed and timing belt pulleys as well as chain sprockets is available as accessory
- A maximum operating distance of 6 meters (20 ft) makes it suitable for use in various applications
- Sturdy aluminium housings provide great assembly stability and accuracy

Pinpoint accuracy with laser technology:

- Aligns grooves of the pulley rather than its face, allowing the alignment of pulleys of unequal width or with dissimilar faces – even fits applications where the pulley face cannot be used as a reference
- No trial and error. The laser position indicates the nature of misalignment allowing easy and accurate adjustment



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Re-lubrication

SKF bearing greases:

The perfect solution for every application

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SKF bearing greases: The perfect solution for every application

Even the very best bearing can only show optimum performance when it is lubricated correctly. Here, it is extremely important to choose the right bearing grease and to apply the most suitable lubrication intervals and methods. This realisation has prompted SKF, the world's leading manufacturer of rolling bearings, to look intensively into the subject of lubrication. SKF engineers consider grease to be a "fundamental" component of the bearing arrangement and thus, as important as the bearing, housing and sealing. SKF's vast experience in the development of rolling bearings forms the basis for the development of a special range of lubricants, the superior quality of which is obtained through continuous testing and studies.

The strict standards and testing parameters developed and applied at the SKF Engineering and Research Centre have become internationally recognised benchmarks for bearing greases. The comprehensive range of SKF bearing greases is the result of many decades of research and development. Each individual lubricant is precisely adjusted to the respective field of application.

SKF sets the standard

Tangible performance parameters mean more to SKF than the chemical composition of the lubricant. The chemical composition is not the only factor in determining the quality of a particular grease, since modern lubricants are extremely complex. SKF has set the standards for developing special testing parameters.

Bearing grease selection

Incorrect lubrication accounts for up to 36 % of premature bearing failures. All-purpose greases are inadequate for specialised bearing needs and can cause problems rather than be beneficial. Bearing applications have wide variations of operating conditions and correct lubrication calls for matching the grease precisely to the application.

Bearing greases help ensure smooth, trouble-free operation and maximum reliability even under the most extreme conditions. They help prevent contamination from penetrating the bearing, cushion any shock loads and protect against corrosion. Selecting the right bearing grease for a certain application is essential for achieving the maximum service life of a bearing.

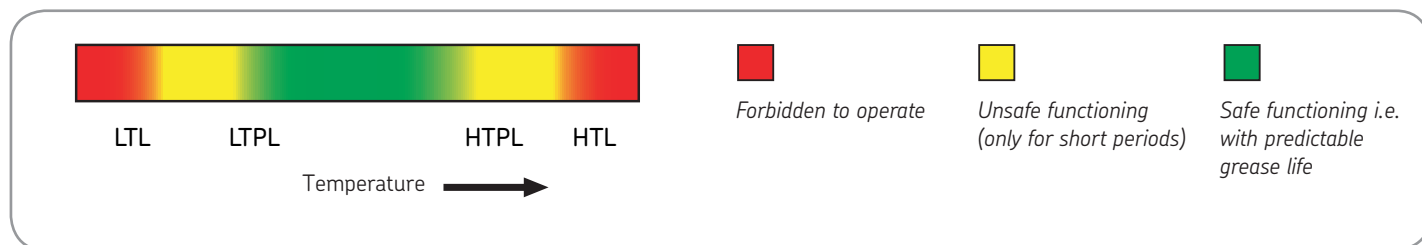
Selection criteria for correct lubrication include bearing type and size, temperatures, speeds and loads, as well as the desired service life and re-lubrication intervals.

SKF Traffic light concept and grease performance factor

The temperature range over which a grease can be used depends largely on the type of base oil and thickener used as well as the additives. The relevant temperatures are schematically illustrated in the following diagrams in the form of a "double traffic light".



Operating temperature range of greases: The SKF Traffic Light Concept



LTL – Low-temperature limit:

The lowest temperature at which the grease will allow the bearing to be started up without difficulty.

LTPL – Low-temperature performance limit:

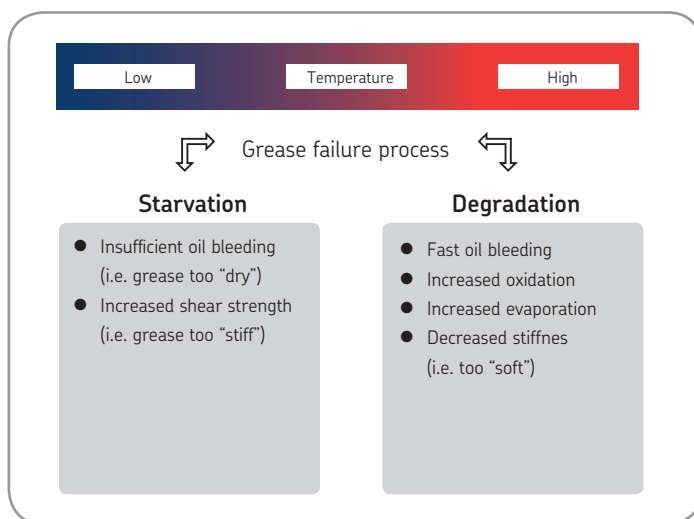
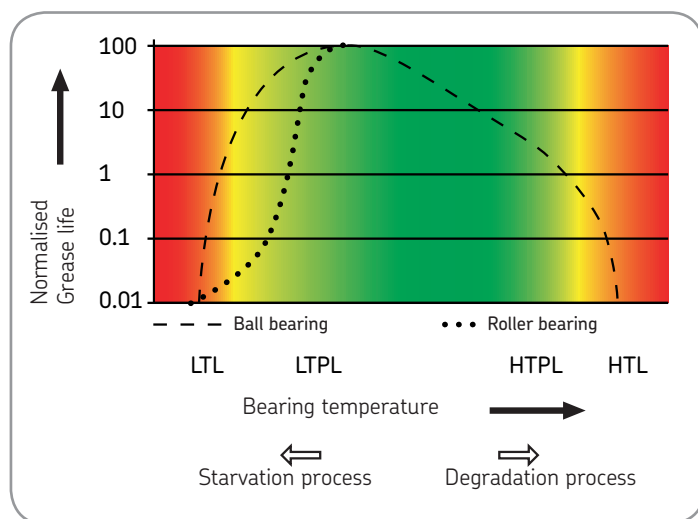
Below this limit, the supply of grease to the contact surfaces of rolling elements and raceways may become insufficient. Values are different for roller and ball bearings.

HTPL – High-temperature performance limit:

Above this limit the grease will oxidise in an uncontrolled way, so that grease life cannot be determined accurately.

HTL – High-temperature limit:

When exceeding this limit, the grease loses its structure permanently (e.g., the dropping point for soap-base greases).



Effect of temperature on grease functioning

The values shown in these diagrams are based on extensive tests conducted in SKF laboratories. These tests allow us to accurately determine the temperature range of the SKF bearing greases given in the selection charts included in this catalogue.

The results of these tests are also used to evaluate the grease life. The performance of each grease is then translated into a Grease Performance Factor (GPF). Greases with highest values offer the longest life. This factor, used in correlation with the SKF re-lubrication diagram (see SKF General Catalogue, allows you to determine the correct re-lubrication intervals for the chosen grease.

Re-lubrication intervals

Choosing the right bearing grease for a certain application is critical to bearing performance. Applying the correct quantity of grease at the right intervals is of equal importance. Over- or under-greasing as well as inadequate lubrication methods can shorten the bearing's service life. For determining the right amount of grease and the correct re-lubrication intervals for a specific application, SKF has developed DialSet, a simple computerised re-lubrication calculation program. Calculated re-lubrication intervals are based on the latest lubrication theories published in the SKF General Catalogue and depend on bearing type used, application conditions and properties of selected bearing grease.



Re-lubrication



Lubrication methods

The lubrication method used is equally important to the right bearing grease, quantity and lubrication intervals. Using lubricators, manual or automatic, facilitates proper lubricant supply to the application. Maintaining cleanliness when lubricating bearings is crucial, as contamination can cause the bearing to fail prematurely.

Using a grease meter in combination with a grease gun or pump during manual lubrication helps ensure the supply of the right quantity of grease. SKF's range of grease guns, pumps and lubrication accessories is designed for contamination-free grease supply as well as ease-of-use.

Continuous lubrication using automatic lubricators, single or multiple-point, provides the application with consistent and controlled supply of bearing grease. This reduces the risk of over- or under-greasing and positively contributes to optimising the bearing's service life. Additionally, automatic re-lubrication reduces the risk of contamination. Around the clock solutions offered by SKF, such as SYSTEM 24 and SYSTEM MultiPoint, provide precise and reliable grease supply, exactly adjusted to the application's needs.

Glossary of lubrication terms

Thickener or soap

Thickener or soap is the system, which holds the oil and/or additives together to enable the lubricating grease to function. The thickener system is formed from either soaps or non-soaps. The type of thickener gives the grease its typical characteristics.

Soaps are based on lithium, calcium, sodium, barium, or aluminium. Non-soaps are based on organic or non-organic solids, bentonite clay, polyurea, silica gel.

Thickener compatibility chart

	Lithium	Calcium	Sodium	Lithium complex	Calcium complex	Sodium complex	Barium complex	Aluminium complex	Clay (Bentonite)	Common polyurea	Calcium sulphonate complex
Lithium	+	○	—	+	—	○	○	—	○	○	+
Calcium	○	+	○	+	—	○	○	—	○	○	+
Sodium	—	○	+	○	○	+	+	—	○	○	—
Lithium complex	+	+	○	+	+	○	○	+	—	—	+
Calcium complex	—	—	○	+	+	○	—	○	○	+	+
Sodium complex	○	○	+	○	○	+	+	—	—	○	○
Barium complex	○	○	+	○	—	+	+	+	○	○	○
Aluminium complex	—	—	—	+	○	—	+	+	—	○	—
Clay (Bentonite)	○	○	○	—	○	—	○	—	+	○	—
Common polyurea	○	○	○	—	+	○	○	○	○	+	+
Calcium sulphonate complex	+	+	—	+	+	○	○	—	—	+	+

+ = Compatible ○ = Test required — = Incompatible

Note: SKF high performance, high temperature bearing grease LGHP 2 is not a common polyurea type grease. It is a di-urea bearing grease, which has successfully been tested for compatibility

with lithium and lithium complex thickened greases i.e. LGHP 2 is compatible with such greases.

Base oil compatibility chart

	Mineral/PAO	Ester	Polyglycol	Silicone: Methyl	Silicone: Phenyl	Polyphenyl-ether	PFPE
Mineral oil / PAO	+	+	—	—	+	○	—
Ester	+	+	+	—	+	○	—
Polyglycol	—	+	+	—	—	—	—
Silicone: methyl	—	—	—	+	+	—	—
Silicone: phenyl	+	+	—	+	+	+	—
Polyphenylether	○	○	—	—	+	+	—
PFPE	—	—	—	—	—	—	+
+ = Compatible ○ = Test required — = Incompatible							

Base oil

The base oil is the oil inside the grease, which provides the lubrication under the operating conditions. Greases are normally based on mineral oils. Synthetic oils can be used for very specific applications such as extremely high or low temperatures. The base oil generally constitutes more than 70% of a grease's composition.

Base oil viscosity

Viscosity is a measure of a fluid's flow characteristics and is usually expressed in terms of the time required for a standard quantity of the fluid, at a given temperature, to flow through a standard orifice. Since viscosity decreases with increasing temperature, the temperature at which it is measured is always stated. The viscosity of base oils is always indicated as a kinematic viscosity abbreviated to cSt, at 40 °C and often also at 100 °C.

Additives

Additives are used to provide additional characteristics such as wear and corrosion protection, friction reducing effects and preventing damage under boundary and mixed lubrication conditions.

Grease consistency/penetration

A measure of the stiffness of a grease. The consistency is classified according to a scale developed by the NLGI (National Lubricating Grease Institute). This is based on the degree of penetration achieved by allowing a standard cone to sink into the grease at a temperature of 25 °C for a period of five seconds. The depth of penetration is measured on a scale of 10^{-1} mm and the softer greases allow the cone to penetrate further into the grease, hence the higher penetration number. The test method is in accordance to DIN ISO 2137.

DIN 51825 Classification system

Bearing greases can be classified according to DIN 51825.

The explanation of the DIN code KP2G-20 is given in the below tables.

Drop point

The drop point is the temperature at which the grease sample, when heated, will begin to flow through an opening and is measured according to DIN ISO 2176. The drop point does not relate to the allowable operating service temperature of the grease.

Mechanical stability

The consistency of a rolling bearing grease should not alter, or only slightly, during the working life of the rolling bearing. Depending on the application, the following tests can be relevant to evaluate the mechanical stability of a grease.

Prolonged penetration

The grease sample is filled into a cup and using an automatic device (called a grease worker) subjected to 100 000 double strokes. At the end of the test the penetration of the grease is measured. The difference between the measured penetration at 60 strokes and after 100 000 strokes penetration is reported as the change in 10^{-1} mm.





Re-lubrication



DIN 51825 – for example: K P 2 G – 20

Application area DIN 51825	K	K= Greases for bearings G= Grease for closed gears OG= Greases for open gears M= Greases for friction bearings/sealing
Additional information	P	P= EP additives F= Solid lubricants E= Ester
NLGI Grade	2	(see NLGI classification)
Upper operating temperature and water resistance	G	(see next table)
Lower operating temperature	-20	-20 °C

Third designation letter

Letter	Upper operating temperature (°C)	Water resistance DIN 51807
C	+60	0 – 40 to 1 – 40
D	+60	2 – 40 to 3 – 40
E	+80	0 – 40 to 1 – 40
F	+80	2 – 40 to 3 – 40
G	+100	0 – 90 to 1 – 90
H	+100	2 – 90 to 3 – 90
K	+120	0 – 90 to 1 – 90
M	+120	2 – 90 to 3 – 90
N	+140	No requirement
P	+160	No requirement
R	+180	No requirement
S	+200	No requirement
T	+220	No requirement
U	>+220	No requirement

Classification of greases by NLGI consistency number

NLGI number	ASTM worked penetration (10 ⁻¹ mm)	Appearance at room temperature	NLGI number	ASTM worked penetration (10 ⁻¹ mm)	Appearance at room temperature
000	445 – 475	very fluid	3	220 – 250	medium hard
00	400 – 430	fluid	4	175 – 205	hard
0	355 – 385	semi-fluid	5	130 – 160	very hard
1	310 – 340	very soft	6	85 – 115	extremely hard
2	265 – 295	soft			

Roll stability

The change in the grease structure (amount of softening or hardening) can be evaluated by filling a cylinder with a pre-specified quantity of grease. A roller is placed inside the cylinder and the complete unit is rotated for 2 hours at room temperature in accordance with ASTM D 1403. SKF modified the standard test procedure to reflect the application conditions under which the grease is used to either 72 or 100 hours at a test temperature of 80 or 100 °C. At the end of the test period the cylinder is allowed to cool to room temperature and the penetration of the grease is measured. The difference between the original penetration and the value measured is reported as the change in penetration in 10⁻¹ mm.

SKF V2F test

The candidate grease is tested for mechanical stability using the following procedure. The test rig consists of a railway axlebox subjected to vibration shocks of 1Hz from a bouncing hammer producing an acceleration level between 12 – 15 g. The test is run at two different speeds, 500 and 1 000 rpm. If the grease, which leaks from the housing through the labyrinth seal which is collected in a tray after 72 hours at 500 rpm, weighs less than 50 grams the test is continued for a further 72 hours at 1 000 rpm. If the total amount of grease leakage after both tests (72 hours at both 500 and 1 000 rpm) does not exceed 150 grams then a rating of 'M' is given. If the grease only fulfils the first part of the test (72 hours at 500 rpm with a grease leakage of 50 grams or less) but fails the second stage, a rating of 'm' is given. If the grease leakage after 72 hours at 500 rpm is greater than 50 grams then it is rated a 'fail'.

Corrosion protection

Lubricating greases should protect metal surfaces from corrosive attack in service. The corrosion protection properties of rolling bearing greases are evaluated using the SKF Emdor method, which is standardised under ISO 11007. Under this test method a mixture of lubricating grease and distilled water is present in the bearing. The bearing alternates during a defined test cycle between standstill and rotation at 80 rpm. At the end of the test cycle the degree of corrosion is evaluated according to a scale between 0 (no corrosion) and 5 (very severe corrosion). A more severe test method is to use salt water to replace the distilled water following the standard test procedure. In addition the test can also be carried out by continuously allowing water to flow or wash through the bearing arrangement during the test cycle.

This test method is called the SKF distilled water washout test. The evaluation procedure is exactly the same as that used under the standardised method. However the procedure places greater demands on the corrosion protection properties of the grease.

Copper corrosion

Lubricating greases should protect copper alloys used in bearings from corrosive attack while in service. The copper corrosion protection properties of rolling bearing greases are evaluated using the standardised method DIN 51811. A copper strip is immersed in the grease sample and placed in an oven. The strip is then cleaned and the degradation is observed. The result is rated by a numerical system.

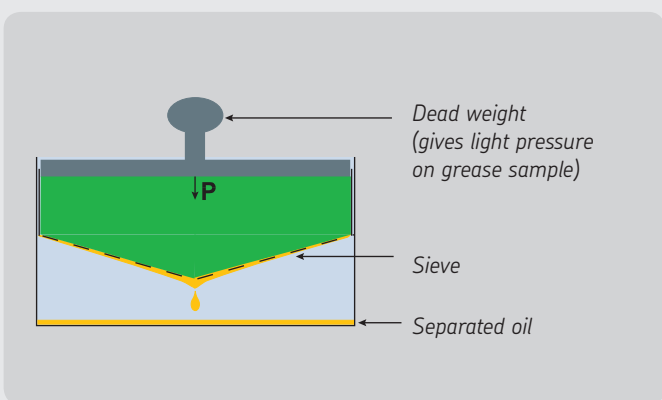
Water resistance

The water resistance of lubricating greases is measured in accordance with DIN 51 807 part 1. A glass strip is coated with the candidate grease, which is placed into a water-filled test tube. The test tube is immersed in a water bath for three hours at a specified test temperature. The change in the grease is evaluated visually and reported as a value between 0 (no change) and 3 (major change) along with the test temperature.

Oil separation

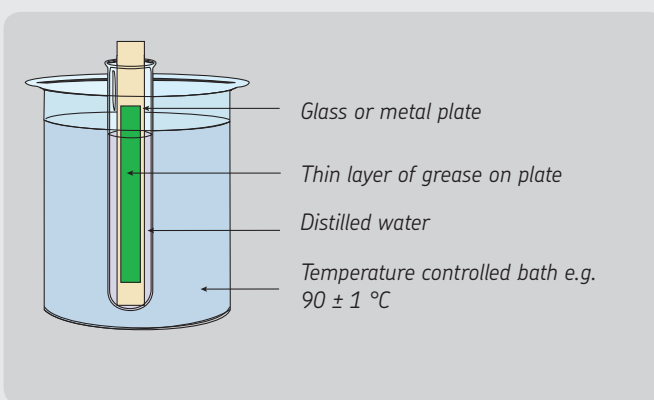
Lubricating greases release oil when stored for long periods of time or when used in bearings as a function of temperature. This phenomenon is necessary to ensure good lubrication. The degree of oil separation will depend upon the thickener, base oil and manufacturing method. A cup is filled with a given quantity of grease (which is weighed before the test) and a 100 gram weight placed on top of the grease. The complete unit is put into an oven at 40 °C for one week. At the end of the week the amount of oil, which has leaked through the sieve is weighed and reported as a percentage of weight loss. The amount of oil separation is measured in accordance with DIN 51 817.

Test for oil bleeding rate



DIN 51 817: determination of the % oil separated after one week at 40 °C

Test for water resistance of greases



DIN 51 807: rating for degree of grease deterioration in water



Re-lubrication



Lubricating ability

The SKF R2F machine assesses the high temperature performance and lubricating ability of a grease, simulating the conditions under which large size bearings operate in housings. The test method is carried out under two different conditions. Test A is conducted at ambient temperature and Test B is conducted at 120 °C. A pass rating in the unheated test (Test A) means that a grease can be used to lubricate larger rolling bearings at normal operating temperatures and also in low vibrating applications. A pass in the heated test (Test B) at 120 °C means that the grease is suitable for use in large roller bearings operating at elevated temperatures.

Rolling bearing grease life

The SKF R0F grease test machine determines the grease life and high temperature performance limit of a lubricating grease. Ten deep groove ball bearings are fitted into 5 housings and filled with a given quantity of grease. The test is undertaken at a pre-determined speed and temperature. Both an axial and radial load is applied and the bearings run until failure. The time to failure is recorded in hours and a Weibull life calculation is made at the end of the test period to establish the grease life. This information can then be used in the determination of re-lubrication intervals in an application.

EP performances

The 4-ball weld load test.

This method evaluates the EP (Extreme Pressure) performance of a lubricating grease. This test method is standardised under DIN 51 350/4. Three steel balls are held in a cup and another fourth ball is rotated against the three balls at a given speed. A starting load is applied and increased at predetermined intervals until the rotating ball siezes and welds to the three stationary balls. The test indicates the point at which the extreme pressure limit of the grease is exceeded. Greases can be considered as EP greases when the weld load is higher than 2600 N.

The 4-ball wear scar test

This test is performed with the same rig used in the 4-ball weld load test. 1400 N are applied on the fourth ball during 1 minute. Then the wear on the three balls is measured. Standard test uses a load of 400 N. However, SKF has decided to increase that to 1400 N in order to make the test relevant for bearing applications.

False Brinelling

Anti-fretting properties of a grease can be relevant for certain applications. SKF can assess these properties using the FAFNIR test standardised as ASTM D4170. Two ball thrust bearings are loaded and oscillated. The wear on each bearing is then measured. Greases offer good fretting protection when the measured wear is below 7 mg.



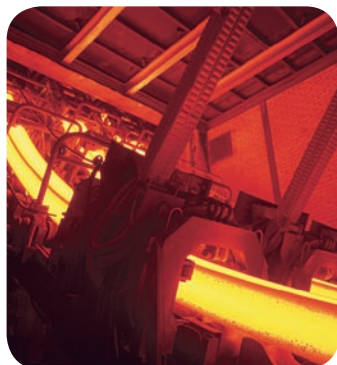
Basic bearing grease selection

Generally use if: Speed = M, Temperature = M and Load = M	LGMT 2	General purpose
Unless:		
Expected bearing temperature continuously > 100 °C / 212 °F	LGHP 2	High temperature
Expected bearing temperature continuously > 150 °C / 302 °F, demands for radiation resistance	LGET 2	Extremely high temperature
Low ambient –50 °C / –58 °F, expected bearing temperature < 50 °C / 122 °F	LGLT 2	Low temperature
Shock loads, heavy loads, frequent start-up / shut-down	LGEP 2	High load
Food processing industry	LGFP 2	Food processing
"Green" biodegradable, demands for low toxicity	LGGB 2	"Green" biodegradable

Note: – For areas with relatively high ambient temperatures, use LGMT 3 instead of LGMT 2
– For special operating conditions, refer to the SKF bearing grease selection chart

Bearing operating parameters

Temperature		
L = Low M = Medium H = High EH = Extremely high	<50 °C / 122 °F 50 to 100 °C / 122 to 230 °F >100 °C / 212 °F > 150 °C / 302 °F	
Speed for ball bearings		
EH = Extremely High VH = Very High H = High M = Medium L = Low	n.dm over 700 000 n.dm up to 700 000 n.dm up to 500 000 n.dm up to 300 000 n.dm below 100 000	
Speed for roller bearings		
H = High M = Medium L = Low VL = Very Low	SRB/TRB/CARB n.dm over 210 000 n.dm up to 210 000 n.dm up to 75 000 n.dm below 30 000	CRB n.dm over 270 000 n.dm up to 270 000 n.dm up to 75 000 n.dm below 30 000
Load		
VH = Very high H = High M = Medium L = Low	C/P < 2 C/P ~ 4 C/P ~ 8 C/P 15	





Re-lubrication



SKF bearing grease selection chart

Bearing working conditions	Temp	Speed	Load	Vertical shaft	Fast outer ring rotation	Oscillating movements	Severe vibrations	Shock load or frequent start-up	Low noise	Low friction
LGMT 2	M	M	L to M	○	—	—	+	—	—	○
LGMT 3	M	M	L to M	+	○	—	+	—	—	○
LGEP 2	M	L to M	H	○	—	○	+	+	—	—
LGFP 2	M	M	L to M	○	—	—	—	—	—	○
LGFB 2	M	M	M to H	○	—	○	+	+	—	—
LGEM 2	M	VL	H to VH	○	—	+	+	+	—	—
LGEV 2	M	VL	H to VH	○	—	+	+	+	—	—
LGLT 2	L to M	M to EH	L	○	—	—	—	○	+	+
LGFL 1	L to M	M to EH	L	—	—	—	○	—	+	+
LGGB 2	L to M	L to M	M to H	○	—	+	+	+	—	○
LGWM 1	L to M	L to M	H	—	—	+	—	+	—	—
LGWA 2	M to H	L to M	L to H	○	○	○	○	+	—	○
LGHB 2	M to H	VL to M	H to VH	○	+	+	+	+	—	—
LGHP 2	M to H	M to H	L to M	+	—	—	+	○	+	○
LGET 2	VH	L to M	H to VH	○	+	+	○	○	—	—

(*1) LTL = Low Temperature Limit
HTPL = High Temperature Performance Limit
(*2) mm²/s at 40 °C / 104 °F = cSt.

(*3) LGGB 2 can withstand peak temperatures of 120 °C / 250 °F
(*4) LGWA 2 can withstand peak temperatures of 220 °C / 428 °F
(*5) LGHB 2 can withstand peak temperatures of 200 °C / 392 °F

Rust inhibiting properties	Description	Temperature range (*1)		Thickener / base oil	Base oil viscosity (*2)
		LTL	HTPL		
+	General purpose industrial and automotive	-30 °C -22 °F	120 °C 250 °F	Lithium soap / mineral oil	110
○	General purpose industrial and automotive	-30 °C -22 °F	120 °C 250 °F	Lithium soap / mineral oil	120
+	Extreme pressure	-20 °C -4 °F	110 °C 230 °F	Lithium soap / mineral oil	200
+	Food compatible	-20 °C -4 °F	110 °C 230 °F	Aluminium complex / medical white oil	130
+	Food compatible, high load	-20 °C -4 °F	120 °C 250 °F	Aluminium complex / PAO and ester	266
+	High viscosity plus solid lubricants	-20 °C -4 °F	120 °C 250 °F	Lithium soap / mineral oil	500
+	Extremely high viscosity with solid lubricants	-10 °C -14 °F	120 °C 250 °F	Lithium-calcium soap / mineral oil	1 020
○	Low temperature, extremely high speed	-50 °C -58 °F	110 °C 230 °F	Lithium soap / PAO oil	18
+	Food compatible, low temperature	-45 °C -49 °F	120 °C 250 °F	Aluminium complex / PAO and ester	110
○	Green biodegradable, low toxicity	-40 °C -40 °F	90 °C (*3) 194 °F	Lithium-calcium soap / synthetic ester oil	110
+	Extreme pressure, low temperature	-30 °C -22 °F	110 °C 230 °F	Lithium soap / mineral oil	200
+	Wide temperature (*4), extreme pressure	-30 °C -22 °F	140 °C 284 °F	Lithium complex soap / mineral oil	185
+	EP high viscosity, high temperature (*5)	-20 °C -4 °F	150 °C 302 °F	Complex calcium sulphonate / mineral oil	400
+	High performance polyurea grease	-40 °C -40 °F	150 °C 302 °F	Di-urea / mineral oil	96
○	Extreme temperature	-40 °C	260 °C	PTFE / synthetic (fluorinated polyether)	400



= Recommended



= Suitable



= Not suitable